Confirming What Matters: Certified Third Normal Form Synthesis from Uncertain Functional Dependencies

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ABSTRACT

State-of-the-art schema design synthesizes an optimal lossless, dependency-preserving Third Normal Form (3NF) decomposition from functional dependencies (FDs). Yet the FDs are mined from data, where most hold only by accident, or suggested by large language models (LLMs), which err. A single false FD can make the design lossy for the true FDs, but a designer cannot be asked about thousands of mined candidates. We propose to certify the design by a proof whose premises are the designer's answers and the data. The designer confirms a cover of the FDs that the design relies on, and refutes each maximal set from which this cover does not derive an attribute, by stating that the set does not determine it. We prove that the price of this certificate, one answer per FD of a smallest cover and one per such set that the data do not refute, is the length of a shortest proof from such answers. Our algorithm Cert3NF lets an advisor built from statistics of the data choose every next question, so that a wrong advisor costs answers but never correctness. On 23 unseen relations, no automatic baseline obtains a good design for more than five, while a simulated designer certifies every design without a budget, at 1.26× to 1.37× the price in the median. In a study with nine people, experts certified all their designs, and practitioners erred on maximal sets about as rarely as on single FDs. In summary, a designer's answers follow the length of the proof, however many FDs hold by accident, and the design stays lossless for the true FDs even where a budget runs out.

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The source code, data, and/or other artifacts have been made available at https://github.com/zhuoxingzhang/cert3nf.

1 INTRODUCTION

The logical design of relational database schemata is a classical topic of database research [12, 36]. Third Normal Form (3NF) synthesis turns a set of functional dependencies (FDs) into a lossless, dependency-preserving decomposition into 3NF [7, 10]. Recent algorithms also optimize the result, for example by minimizing how many minimal keys and non-key FDs the schemata must maintain [55–57]. These guarantees are relative to the input FDs.

Yet all approaches so far assume that the input is exactly the set Σ of business rules of the domain. In practice, Σ is estimated. FD discovery returns every minimal FD that holds on an instance [24, 41], and most of these hold by accident [50]. On the eight tables of TPC-H [46], for example, discovery returns 3,945 minimal FDs, of which only 76 follow from the specification. Large language models (LLMs) that judge FDs from the names of the attributes err frequently on real schemata. Hence, an estimate $\hat{\Sigma}$ differs from Σ, and synthesis computes the optimal design for the wrong input. A designer who knows the domain can tell genuine rules from accidental FDs, but cannot be asked about thousands of candidates. This is illustrated as follows.

Example 1.1 (Running example). Consider the schema Orders with the attributes O(rder), customer N(umber), C(ity) and D(ate). Its business rules are $\Sigma = \{O \to N, O \to D, N \to C\}$: an order has one customer and one date, and a customer lives in one city. Orders violates 3NF under these rules, since $N \to C$ holds while N is not a key. On these rules, iCONF [57], the state of the art in 3NF synthesis, returns the decomposition $\{ON, OD, NC\}$. On the following sample, FD discovery also reports $N \to D$, since every customer ordered on one date only.

Order	Cust. No.	City	Date
o_1	101	Auckland	3 March
o_2	101	Auckland	3 March
o_3	102	Wellington	3 March
o_4	103	Auckland	5 March
o_5	104	Wellington	3 March

On the estimate $\hat{\Sigma} = \Sigma \cup \{N \to D\}$, synthesis returns $\hat{\mathcal{D}} = \{ON, NC, ND\}$, which declares N as the key of ND. This design rejects legal data, such as a second order of customer 101 on another date. Synthesis adds no key schema, since ON contains the key O. The design is lossless, dependency-preserving and in 3NF for $\hat{\Sigma}$, as synthesis guarantees for any input. Yet it is lossy for Σ: joining ON with ND assigns to every order of a customer the dates of all orders of this customer. In contrast, if the estimate misses $O \to D$, synthesis still returns $\{ON, NC, OD\}$, now with the key schema OD, which does not declare O as a key. Declaring O as the key of OD repairs the design without moving any data. □

Ex. 1.1 raises four questions. Firstly, errors differ in their harm. The false FD $N \to D$ broke losslessness for Σ, so its repair moves data, while the missing FD $O \to D$ only left a key undeclared,

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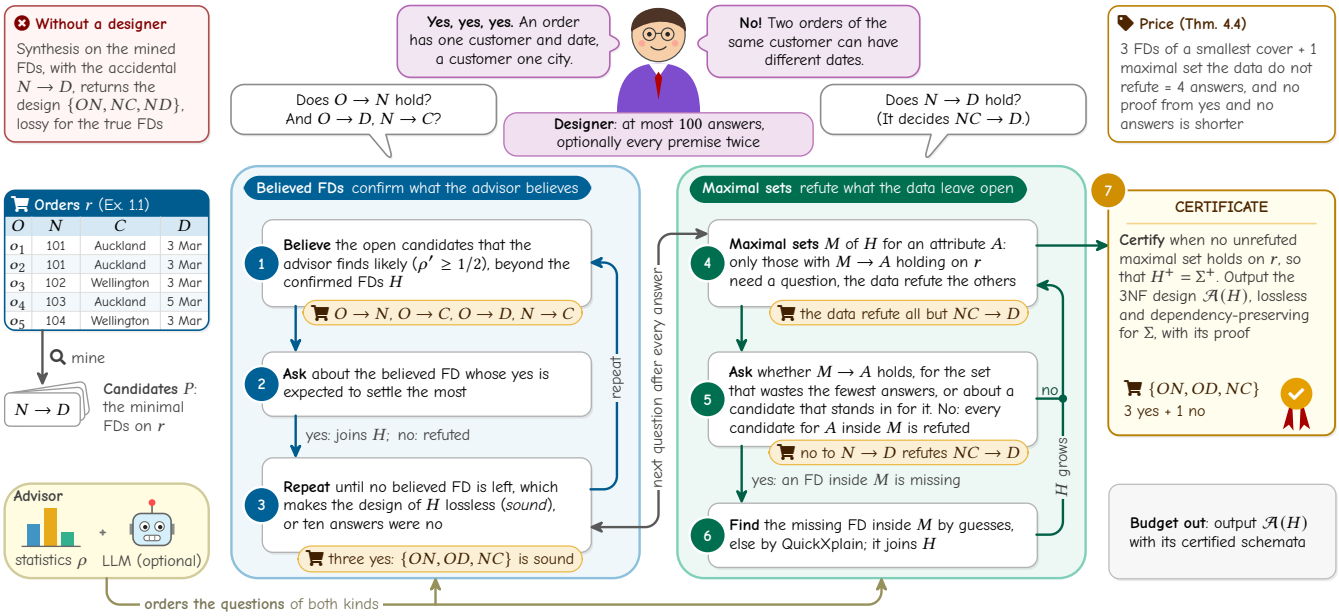


Figure 1: Cert3NF on Ex. 1.1: Confirm What the Advisor Believes (1–3), Refute What the Data Leave Open (4–6)

which a declaration repairs after deployment. Hence, we ask (Q1) *Which errors of an estimate must be resolved before deployment?*

Secondly, once the designer has confirmed the three rules of Σ, the design $\{ON, OD, NC\}$ is lossless for the true FDs, whatever else holds. Whether it preserves every rule depends on what else holds, such as $N \to D$. The sample refutes every further FD except $N \to D$ and $NC \to D$, so whether any further rule holds is one question: can two orders of the same customer have different dates? If they can, the three confirmed rules are all there is. This leads to (Q2) *Which answers certify a design, and how many are needed?*

Thirdly, advice can anticipate the answer about dates: the statistics of the sample doubt $N \to D$, and an LLM knows from the attribute names that a customer does not determine a date. Such advice is right here, but errs on many other candidates. This raises (Q3) *What can advice that may be wrong contribute to a certificate?*

Finally, designers err, and a single wrong answer can certify a wrong design. We therefore ask (Q4) *Does a certificate survive a designer who errs?*

We answer these questions for iCONF and every synthesis algorithm that differs from it only in the order in which it eliminates redundant FDs. Our main idea is to certify the design by a proof whose premises are the designer's answers and the records of the data. This proof, the *certificate*, shows that the confirmed FDs imply exactly the true FDs, so that the guarantees of synthesis for its input become guarantees for the true FDs. It has two parts. The designer confirms a cover of the FDs that the design relies on, which makes the design lossless for the true FDs. The designer and the data also refute every *maximal set* of the confirmed FDs, which makes the design dependency-preserving and in 3NF for the true FDs as well. Such a set is maximal among the sets of attributes from which the confirmed FDs do not derive a given attribute. Refuting it is the kind of answer that Armstrong relations and design by example ask for [6, 38]: two records can agree on these attributes and differ on that one. One such answer settles every candidate for that attribute whose left-hand side lies inside the set. The data refute some of these sets for free. On the join of the lineitem and orders tables of TPC-H, for example, a certificate needs only 49 answers among 36,883 candidates, and our algorithm finds one with 54.

Fig. 1 shows the resulting algorithm Cert3NF on our running example. It mines the candidates, and an *advisor* attaches to each a probability from the statistics of the data. After every answer, Cert3NF asks the designer to confirm an FD that the advisor believes or, once none is left, to refute the maximal set that wastes the fewest answers in expectation. Finally, it synthesizes the certified design from the confirmed FDs. That is, the advisor decides the order of the questions, and the designer decides the design. Our **contributions** are as follows.

(1) Conceptually, we grade the damage of an estimate by the cost of its repair (Sec. 3, Q1). Only false FDs can make a design lossy for the true FDs, and a single one does so in one of two ways: by fabricating a key or by short-circuiting true FDs. Missing FDs, in contrast, always admit illegal database states and can cost preservation or 3NF. Every candidate decides the constraints that the design declares, so a certificate has to settle every candidate, and it can only afford to do so in groups.
(2) Theoretically, we show that a confirmed cover and refuted maximal sets certify a design exactly (Sec. 4, Q2). The *price* of the certificate is one answer per FD of a smallest cover and one per maximal set of the true FDs that the data do not refute, however many FDs hold by accident. No proof of the true FDs from yes and no answers is shorter.
(3) Algorithmically, we propose Cert3NF (Fig. 1, Sec. 5). Beyond one yes answer per confirmed FD, a wrong advisor costs at most ten no answers, one wrong guess per refuted set and a

logarithmic number of answers per FD it fails to predict, but never the correctness of the certificate (Q3). No method that does not know the true FDs avoids the logarithmic term. On data that violate true FDs in a few records, a tolerance keeps the certificate exact (Q2). For a designer who errs, every premise of the proof is answered twice, which bounds the probability of a wrong certificate (Q4).

(4) Empirically, we evaluate Cert3NF by rules fixed in advance, on 23 confirmatory relations never used during development (Sec. 6). No fully automatic method that we tested, including Normalize, an LLM alone and a multi-agent normalizer, obtains a good design for more than five of them, while a simulated designer certifies at least 19 within a hundred answers and all without a budget, at 1.26× to 1.37× the price in the median. On synthetic joins with up to a million candidates, the answers follow the price rather than the candidates. Attribute exploration needs 7× to 9× our answers, and EC^2, even within our framework, spends 2× to 3× our answers above the price and certifies half as many designs within fifty answers. Where the budget runs out, the design of Cert3NF is still lossless, while up to a third of those of the interactive baselines are lossy (Q2). The statistics of the data alone provide what the advisor needs, so an LLM stays optional (Q3). Within the budget, answering every premise twice cuts the wrong certificates from more than one half to about 2% for a designer who errs independently on up to one answer in twenty, and in a study with nine people, it caught the slips of practitioners, who erred about as rarely on maximal sets as on single FDs (Q4).

(5) Practically, Cert3NF runs online at *https://cert3nf.onrender.com* for practitioners and researchers to design their own tables.

In summary, we establish the first approach to schema design that prices the certificate of a design against the unknown true FDs.

Organization. Sec. 2 recalls 3NF synthesis, Sec. 3 grades errors by the cost of their repair, Sec. 4 defines the certificate and proves its price, Sec. 5 develops Cert3NF, and Sec. 6 presents our experiments. We discuss related work in Sec. 7 and conclude in Sec. 8. Appendix A contains the proofs and Appendix B the details of the experiments. Both are also in the full version of this paper, `Cert3NF-full.pdf`, in the repository of our code, online demo and results at *https://github.com/zhuoxingzhang/cert3nf*.

2 FUNDAMENTALS AND PREVIOUS WORK

We recall the notions of schema design to which the guarantees of synthesis refer, and introduce the elimination skeleton that the algorithms we study share. Ex. 1.1 illustrates them.

Dependencies. A relation schema is a finite set R of attributes. A functional dependency (FD) over R is an expression $X \to A$ with $X \subseteq R$ and $A \in R$, which a relation over R satisfies if any two of its records that agree on X also agree on A. For a set Σ of FDs, Σ^+ denotes the set of FDs that Σ implies [4], and $X^+_\Sigma = \{A : X \to A \in \Sigma^+\}$ the closure of X. The *agree set* of two records is the set of attributes on which they agree, and an *Armstrong relation* for Σ satisfies exactly the FDs in Σ^+ [4, 6]. The *atomic closure* Σ_a, which depends only on Σ^+, consists of the FDs $X \to A \in \Sigma^+$ with $A \notin X$ such that $Y \to A \notin \Sigma^+$ for every $Y \subsetneq X$. For $S \subseteq R$, the projection $\Sigma[S] = \{X \to A \in \Sigma_a : XA \subseteq S\}$ is a cover of the FDs in Σ^+ whose attributes lie within S. A key of (R, Σ) is a minimal $K \subseteq R$ with $K^+_\Sigma = R$, a superkey is a superset of a key, and an attribute is prime if it belongs to some key. A schema S is in 3NF with respect to Σ if, for every $X \to A \in \Sigma[S]$, X is a superkey of $(S, \Sigma[S])$ or A is prime for $(S, \Sigma[S])$. In Ex. 1.1, the projection of $\hat{\Sigma}$ onto ND is $\{N \to D\}$, while that of Σ is empty.

Designs. A design $\mathcal{D}$ over R is a decomposition of R, that is, a set of schemata whose union is R. Under a set Σ of FDs, each schema $S \in \mathcal{D}$ declares the constraints $\Sigma[S]$. The design is lossless with respect to Σ if every relation over R that satisfies Σ is the natural join of its projections onto the schemata of $\mathcal{D}$. It preserves Σ if $\bigcup_{S \in \mathcal{D}} \Sigma[S]$ implies Σ. We call $\mathcal{D}$ *Σ-valid* if it is lossless and dependency-preserving with respect to Σ and each of its schemata is in 3NF with respect to Σ. The following classical sufficient condition underlies 3NF synthesis.

Lemma 2.1 (Embedded cover [10]). *If a design $\mathcal{D}$ embeds a cover F of Σ, that is, each $X \to A \in F$ has $XA \subseteq S$ for some $S \in \mathcal{D}$, and some schema of $\mathcal{D}$ contains a key of (R, Σ), then $\mathcal{D}$ is lossless and dependency-preserving with respect to Σ.*

In Ex. 1.1, the design $\{ON, OD, NC\}$ embeds Σ and contains the key O, so it is lossless and preserves Σ.

The elimination skeleton. The synthesis algorithms we study differ only in the order in which they eliminate redundant FDs. This motivates the following definition.

Definition 2.2 (Elimination skeleton). Given Σ and an order π on Σ_a, the algorithm $\mathcal{A}_\pi$ starts with $G = \Sigma_a$ and walks Σ_a in the order π. It removes each FD σ from G that $G \setminus \{\sigma\}$ implies, and keeps the schema XA of every FD $X \to A$ that it does not remove. The final G is a non-redundant cover of Σ. The output $\mathcal{A}_\pi(\Sigma)$ consists of the maximal kept schemata, plus a minimum-size key of (R, Σ), chosen by a fixed order, if no kept schema contains a key.

That is, the skeleton keeps one schema per FD of a non-redundant cover, and the order decides which cover it keeps. iCONF [57] first walks the FDs whose schemata XA violate BCNF, by decreasing number of non-key FDs of XA, and then the others by decreasing number of minimal keys of XA. The classical order walks Σ_a by increasing size of the left-hand side and then lexicographically in the order of the columns. The next result shows that every order yields a Σ-valid design, and that together the orders reach every non-redundant cover in the atomic closure.

Lemma 2.3 (Skeleton guarantees). *Let the order π depend only on Σ_a, with ties broken by a fixed order of the attributes. Then $\mathcal{A}_\pi(\Sigma)$ depends only on Σ^+ and is Σ-valid. Over all orders, the covers G that $\mathcal{A}_\pi$ keeps are exactly the non-redundant covers of Σ contained in Σ_a.*

Hence, every member of the family inherits the guarantees of 3NF synthesis for the FD set it is given. What these guarantees mean for the true FDs when the given set is wrong, and how a designer can certify a design against the true FDs, have not been considered before. This is the contribution of our current work.

3 ERRORS AND THEIR DAMAGE

Synthesis runs on an estimate $\hat{\Sigma}$, while the domain imposes the true set Σ. By Lem. 2.3, the design is always $\hat{\Sigma}$-valid. Hence, from now on, the terms “lossless”, “lossy”, “dependency-preserving” and

"in 3NF" refer to Σ unless another FD set is named. We define the errors of $\hat{\Sigma}$ and their damage to the design, grade the damage by the cost of its repair, and show which errors a certificate must settle. We start with single errors.

Definition 3.1 (Single errors). An estimate $\hat{\Sigma} = \Sigma \cup \{\sigma\}$ with $\sigma \notin \Sigma^+$ has the single *false FD* σ. Given a non-redundant cover G of Σ, an estimate $\hat{\Sigma} = G \setminus \{\sigma\}$ with $\sigma \in G$ has the single *missing FD* σ.

In general, the false FDs of $\hat{\Sigma}$ are those in $\hat{\Sigma}^+ \setminus \Sigma^+$, and its missing FDs those in $\Sigma^+ \setminus \hat{\Sigma}^+$. In Ex. 1.1, $N \to D$ is a single false FD and $O \to D$ a single missing FD. Errors damage a design at two levels, the declared constraints and the schemata.

Definition 3.2 (Damage). Let $\hat{\mathcal{D}}$ be synthesized from $\hat{\Sigma}$, so that each schema S declares $\hat{\Sigma}[S]$. Relative to Σ, the design *over-enforces* if some declared FD is not in Σ^+, and *under-enforces locally* if $\Sigma[S] \not\subseteq (\hat{\Sigma}[S])^+$ for some schema S. It suffers *structural damage* if it is not Σ-valid.

The two levels differ in the cost of their repair. Over- and under-enforcement are repaired by declaring $\Sigma[S]$ instead of $\hat{\Sigma}[S]$ on the existing tables, which moves no data, while structural damage requires a new decomposition and a migration of the data. In Ex. 1.1, the design $\{ON, NC, ND\}$ over-enforces $N \to D$ and suffers structural damage, since it is lossy for Σ and does not preserve $O \to D$. In contrast, the design for the missing FD $O \to D$ is Σ-valid and only under-enforces locally. The direction of an error determines which damage at the level of constraints it causes, and only false FDs can cause lossiness, as the next result shows.

Proposition 3.3 (Asymmetry). *(a) If $\hat{\Sigma} \subseteq \Sigma^+$, every design that is lossless with respect to $\hat{\Sigma}$ is lossless with respect to Σ.*

(b) A design that preserves $\hat{\Sigma}$ over-enforces if and only if $\hat{\Sigma} \not\subseteq \Sigma^+$, and it then rejects a relation that satisfies Σ.

(c) If $\hat{\Sigma}^+ \subsetneq \Sigma^+$, every design that is lossless with respect to $\hat{\Sigma}$ accepts a database state that is not the projection of any relation satisfying Σ.

Since synthesized designs are $\hat{\Sigma}$-valid, Prop. 3.3 applies to them: a false FD always rejects legal data, while a missing FD never causes lossiness but always admits illegal states. In Ex. 1.1, the false FD $N \to D$ rejects the second order of customer 101, and the missing FD $O \to D$ admits two dates for one order.

When a false FD breaks losslessness. Lossiness is the damage that moves data, and we characterize how a single false FD causes it. From now on, $\sigma = X \to a$. Let $\Sigma[\hat{\mathcal{D}}] = \bigcup_{S \in \hat{\mathcal{D}}} \Sigma[S]$ be the true FDs embedded in $\hat{\mathcal{D}}$. The following theorem separates the two ways in which lossiness arises: the design relies on a key that the false FD fabricated (*key loss*), or it contains a true superkey, but the false FD short-circuits the true FDs that would carry it to every attribute (*path loss*).

Theorem 3.4 (Lossiness). *Let $\hat{\Sigma} = \Sigma \cup \{\sigma\}$ have the single false FD σ, and let $\hat{\mathcal{D}} = \mathcal{A}_\pi(\hat{\Sigma})$ for any order π.*

(a) (Key loss) No schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ if and only if $a \notin S^+_\Sigma$ for every schema $S \in \hat{\mathcal{D}}$ with $S^+_{\hat{\Sigma}} = R$. In this case $\hat{\mathcal{D}}$ is lossy, and some minimal key of $(R, \hat{\Sigma})$ is not a superkey with respect to Σ.

(b) (Path loss) If some schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ but $\hat{\mathcal{D}}$ is lossy, then $\hat{\mathcal{D}}$ does not preserve Σ, and every FD $Y \to B \in \Sigma^+$ that $\Sigma[\hat{\mathcal{D}}]$ does not imply satisfies $X \subseteq Y^+_\Sigma$.

Ex. 3.5 shows key loss.

Example 3.5 (Key loss). Let $R = ABCD$ and $\Sigma = \{C \to D, BC \to A\}$, with the single key BC. The false FD $D \to C$ makes BD a key as well. Both iCONF and the classical order walk $BC \to A$ before $BD \to A$ and remove it, since $C \to D$ and $BD \to A$ imply it, which yields $\{ABD, CD\}$, whose schema ABD contains the key BD. Since $(ABD)^+_\Sigma = ABD$ does not contain C, no schema is a superkey under Σ. Hence, the design is lossy. With the attributes in reverse order, both walk $BD \to A$ first and return $\{ABC, CD\}$, which is Σ-valid, so the order of the attributes decides whether this false FD breaks losslessness. □

Ex. 1.1 shows path loss. The false FD $N \to D$ makes $O \to D$ redundant, so synthesis keeps the cover $\{O \to N, N \to C, N \to D\}$. The schema ON is a superkey under Σ, yet $\Sigma[\hat{\mathcal{D}}] = \{O \to N, N \to C\}$ carries it only to ONC. The missing attribute D is reached only through $O \to D$, which the false FD short-circuits, as $X = N \subseteq O^+_\Sigma$. Sec. 6.2 shows that nine in ten accidental FDs on TPC-H make the design lossy, all by path loss.

Every candidate decides the constraints. Damage at the level of constraints is cheap to repair, but it cannot be ruled out without settling every candidate. We call two designs *constraint-equivalent* if they have the same schemata with equivalent declared constraints.

Proposition 3.6 (Constraint equivalence). *Let $\mathcal{A}$ be an algorithm of the family. If $\mathcal{A}(\Sigma')$ and $\mathcal{A}(\Sigma'')$ are constraint-equivalent for sets Σ' and Σ'' of FDs, then $\Sigma'^+ = \Sigma''^+$.*

That is, two candidate sets that differ in their closure synthesize designs that differ in their constraints. Hence, whether a design declares the right constraints depends on every candidate, however unlikely. In Ex. 1.1, $O \to D$ only decides whether OD declares the key O, yet a certificate must still settle it.

In summary, the answer to (Q1) is that every candidate must be settled before deployment: a false FD can make the design lossy, a missing FD can cost preservation or 3NF (Sec. 6.2), and every candidate decides the constraints that the design declares. Hence, a certificate has to show that the design is lossless, which Thm. 3.4 ties to the FDs the design relies on, and that it declares the true constraints, which Prop. 3.6 ties to every candidate. The next section shows that the answers of a designer establish both and settle the candidates in groups.

4 CERTIFICATES FROM ANSWERS

We now define the certificate that Cert3NF issues, and show what it costs. The designer answers one kind of question, whether an FD $Z \to A$ holds in the domain, for any $Z \subseteq R$ and $A \in R$. That is, the designer decides whether two records that agree on Z must agree on A.

Setting. The *known* FDs Σ_0 are true FDs given before any question, such as the FDs of a declared key, and Σ_0 may be empty. The candidates P are the minimal FDs that hold on the data and that Σ_0 does not imply, and we assume, as FD discovery does, that the data satisfy Σ. The designer's answers divide into the FDs Σ_y answered

yes and the FDs answered no, and $H = \Sigma_0 \cup \Sigma_y$ holds everything confirmed. A no to $Z \to B$ refutes every FD $X \to A$ with $X \subseteq Z_H^+$ and $B \in (ZA)_H^+$, since $X \to A$ and H would imply $Z \to B$. A candidate is *open* if H does not imply it and it is not refuted. We write $\mathcal{A}$ for the synthesis algorithm in use. We call a design $\mathcal{D}$ *good* for a set Σ' of FDs if it is Σ'-valid and $|\mathcal{D}| \leq |\mathcal{A}(\Sigma')|$. That is, a good design has every guarantee of synthesis for Σ' and no more schemata than synthesis needs.

Soundness and completeness. A certificate has to show that the design is good for Σ, which is unknown, from H, which is known. Confirmed FDs rule out false FDs, which gives soundness, and refuted maximal sets rule out missing FDs, which gives completeness. We assume for now that the designer answers truthfully, so $H \subseteq \Sigma^+$. Hence, a design that is good for H is lossless with respect to Σ by Prop. 3.3(a), and the tables it keeps are justified by confirmed FDs alone. We call such a design *sound*. What soundness leaves open is completeness: a true FD that H does not imply may be unpreserved, violate 3NF inside a schema, or admit a smaller design. Completeness is decided by the maximal sets of attributes from which H does not derive a given attribute. This motivates the following definition.

Definition 4.1 (Maximal sets). For a set H of FDs over R and an attribute $A \in R$, the *maximal sets* $\max(H, A)$ are the maximal $M \subseteq R$ with $A \notin M_H^+$.

Thus, H implies $X \to A$ if and only if X lies within no member of $\max(H, A)$. These are the maximal sets of Mannila and Räihä, the agree sets of an Armstrong relation for H that are maximal among those not containing A [6, 38]. In Ex. 1.1, for $H = \Sigma$, the maximal set for D is NC: two orders of the same customer, who has one city, may still differ in the date, and so may any two orders that agree on less. Since the maximal sets determine the closure of an FD set, they also decide when confirmed FDs are complete, as the following theorem shows.

Theorem 4.2 (Certificate). *Let $H \subseteq \Sigma^+$, and let $\mathcal{D}$ be good for H. Then $H^+ = \Sigma^+$ if and only if $M \to A \notin \Sigma^+$ for every $A \in R$ and every $M \in \max(H, A)$. In this case, $\mathcal{D}$ is good for Σ, and $H[S] = \Sigma[S]$ for every schema $S \in \mathcal{D}$, so that every schema declares the true constraints.*

By Thm. 4.2, completeness comes down to one question per maximal set $M \in \max(H, A)$, namely whether $M \to A$ holds in the domain. The certificate thus consists of yes answers that justify the design and no answers that refute the maximal sets that hold on the data. No estimate enters it. We illustrate the theorem on our running example.

Example 4.3 (Certificate for Orders). Consider again Ex. 1.1 with $\Sigma_0 = \emptyset$. Once the designer has confirmed $O \to N$, $O \to D$ and $N \to C$, the design $\{ON, OD, NC\}$ is good for $H = \Sigma$, so it is sound. The maximal sets are NCD for O, CD for N, D for C and NC for D. The sample refutes three of them: o_1 and o_2 agree on NCD and differ on O, o_3 and o_5 agree on CD and differ on N, and o_1 and o_3 agree on D and differ on C. Only $NC \to D$ holds on the sample, and the designer answers that two orders of the same customer can have different dates. Hence, $H^+ = \Sigma^+$, and the design is certified by three yes answers and one no answer. □

The example shows that the data can refute maximal sets without a question, and that the remaining ones are exactly those that hold on the data. For a relation r that satisfies Σ, a set $M \subseteq R$ is *held* by r if $M \in \max(\Sigma, A)$ and $M \to A$ holds on r for some attribute A, that is, if no two records of r refute $M \to A$. We write m_r for the number of held sets, where a set that is maximal for several attributes counts once. Every agree set of r is closed under Σ, so the FD $M \to A$ with $M \in \max(\Sigma, A)$ fails on r exactly when M is the agree set of two records of r. A set is thus held for all attributes for which it is maximal or for none. Hence, m_r counts the maximal sets of Σ that are not agree sets of r, and it is zero exactly when r is an Armstrong relation for Σ [6].

A set of answers *certifies* a design if the design is good for H and every $M \in \max(H, A)$ with $M \to A$ holding on r is refuted by a no answer. The design is then good for Σ by Thm. 4.2. The next result makes the example precise: a certificate needs one yes answer per FD of a cover and one no answer per held set, and no fewer.

Theorem 4.4 (Price of a certificate). *Let r satisfy Σ, and let F be a set of FDs with $(\Sigma_0 \cup F)^+ = \Sigma^+$.*

- *(a) For every $H \supseteq \Sigma_0$ and $M \in \max(H, A)$, the FD $M \to A$ holds on r if and only if M contains the left-hand side of a candidate $X \to A$.*
- *(b) Yes answers to the FDs of F and, for every held set M, a no answer to one FD $M \to A$ with $M \in \max(\Sigma, A)$ that holds on r certify every design that is good for Σ, with $|F| + m_r$ answers in all.*
- *(c) Every set of answers that certifies a design contains at least m_r no answers, and its yes answers Σ_y satisfy $(\Sigma_0 \cup \Sigma_y)^+ = \Sigma^+$.*
- *(d) A set of answers certifies a design that is good for Σ if and only if Σ^+ is the only closed set of FDs that holds on r, contains Σ_0 and the yes answers, and contains no FD answered no.*

We call $|F| + m_r$ for a smallest F the *price* of a certificate. Thm. 4.4 shows that the price is set by the domain and the data, not by the number of candidates. Indeed, the yes answers are the cover that a designer would write down anyway, and the no answers are one per maximal set that the data do not refute. One no settles at once every candidate that leads from inside the set to an attribute outside it. By (b) and (c), the price is the length of a shortest proof of the true FDs from yes and no answers. By (d), it is also the size of a smallest teaching set [17, 22] for Σ^+ among the closed sets of FDs that contain Σ_0 and hold on the data. Unlike the teaching dimension, it depends on the data. A smallest F is a minimum Horn base, which is NP-hard to find [20], but integer programming finds it in practice (Appendix B).

Both theorems refine classical facts. Thm. 4.2 carries over to designs that a closure system is determined by its meet-irreducible sets, which are the maximal sets of design by example and the characteristic models of the Horn theory of the FDs [28, 38]. Without data, the certificate of Thm. 4.4(b) is a cover together with one excluded FD per maximal set [19, 25]. With data, only the held sets need an answer, and the more maximal sets the data witness, the lower the price.

Part (a) also shows why the certificate refutes maximal sets rather than candidates. A true FD $Y \to A$ whose left-hand side is not minimal on the data is not a candidate, since some $X \subsetneq Y$ determines A on the data by accident. Refuting the candidate $X \to A$ would leave $Y \to A$ unsettled. In contrast, while H does not

Algorithm 1 Cert3NF$(r, \Sigma_0, P, \rho, B, b)$ in Its Trusting Mode

Require: Data r over R, known FDs Σ_0, candidates P, an estimator ρ, the budget B and the cap b
Ensure: $\mathcal{A}(H)$ with its certificate, or, after B answers, the sound design $\mathcal{A}(H)$ with its certified schemata

```
1: H ← Σ0; nos ← 0
2: while a pair (M, A) with M ∈ max(H, A) needs a question do
3:     (M, A) ← such a pair with the smallest p_M(1 + e_M)
4:     if an open candidate X → A has X+_H = M then
5:         Ask(X → A) for the likeliest such X            ▷ in place of M → A
6:     else if nos < b and the advisor believes an open candidate then
7:         c ← the believed FD with the largest ρ'_c(1 + n_c)
8:         if not Ask(c) then
9:             nos ← nos + 1
10:    else
11:        if p̄(1 + h') > 1 and a guess X → A inside M is likely then
12:            Ask(X → A)                                 ▷ the first guess before its set
13:        if M ∈ max(H, A) then                          ▷ unless the guess hit
14:            if Ask(M → A) then
15:                H ← H ∪ {Find(M, A)}
16:    update ρ' by the answers and recompute the maximal sets of H
17:    if B answers were given then
18:        return A(H) with its certified schemata
19: return A(H) and its certificate
```

imply $Y \to A$, some $M \in \max(H, A)$ contains Y. The FD $M \to A$ then holds on the data by (a) and is true, and the designer's yes to it reveals a missing FD inside M.

The maximal sets can be exponentially many [6], but the price counts only the held ones. Thm. 4.4 assumes that the right questions are asked, and the next section shows how to find them.

5 THE ALGORITHM CERT3NF

Cert3NF has to find the cover and the held sets of Σ without knowing Σ, within a budget of B answers, which we set to 100. The designer confirms the cover of the design that an *advisor* believes, and refutes the maximal sets of the confirmed FDs, as in Fig. 1. Alg. 1 shows the loop in the *trusting mode*, which accepts every answer. Each iteration takes the pair (M, A) that wastes the fewest answers in expectation (line 3) and asks about a candidate that stands in for $M \to A$ (lines 4–5), else about a believed FD (lines 6–9), else about $M \to A$ itself (lines 10–15). The advisor is any source of a probability per candidate, and ours is built from the statistics of the data alone. The advisor only orders the questions, and we bound what a wrong advisor can cost.

The advisor. An estimator assigns each candidate $c \in P$ a probability ρ_c from statistics of the data, such as the size of its left-hand side, the redundancy it causes and the numbers of distinct values, by a logistic model trained on other schemata. Such estimators rank candidates well but are rarely calibrated on a new schema, so the advisor recalibrates them by a prior learned on other schemata and updated by the designer's answers. Candidates that share a left-hand side are not independent, since a left-hand side X that determines one attribute is likely to determine more. The advisor therefore models whether each X is a determinant, so that a yes on a candidate of X raises the other candidates of X and a no lowers them (Appendix B). The result is the probability ρ'_c of the advisor. An LLM may add evidence through the likelihood ratio of its answer, and it is asked only where its answer can change the next question. Sec. 6.4 shows that the statistics alone do nearly as well.

Believed FDs. The advisor *believes* the open candidates with $\rho'_c \geq 1/2$. These FDs and the confirmed FDs $H = \Sigma_0 \cup \Sigma_y$ form its estimate $\hat{\Sigma}$, and the *target design* is $\mathcal{A}(\hat{\Sigma})$. Cert3NF asks about the believed FD c with the largest $\rho'_c(1 + n_c)$ (line 7), where n_c is the number of other believed FDs that $H \cup \{c\}$ implies. A yes to c settles these FDs without an answer of their own. As with every question that Ask poses, a yes puts the FD into H, a no refutes it and every candidate that would imply it with H (Sec. 4), and the advisor and $\hat{\Sigma}$ are recomputed (line 16). Once no believed FD is left, $\hat{\Sigma}^+ = H^+$ and the target design is good for H and hence sound. An advisor that insists on false FDs costs at most b no answers, with $b = 10$, before Cert3NF stops asking about believed FDs (line 6). An advisor that believes true FDs beyond a smallest cover costs yes answers instead, which Prop. 5.2 counts.

Maximal sets. By Thm. 4.4(a), a pair (M, A) with $M \in \max(H, A)$ needs a question only if M contains the left-hand side of a candidate of A. Otherwise, two records of r refute it without a question. A candidate $X \to A$ with $X^+_H = M$ stands in for $M \to A$, since its no refutes $M \to A$ as well and its yes yields the FD with one answer instead of two, so Cert3NF asks such a candidate first, even before the believed FDs (lines 4–5). A no answer to a held set is part of every proof, so only a yes can waste answers. Hence, Cert3NF asks about the pair with the smallest expected waste $p_M(1 + e_M)$ (line 3). Here, $p_M = 1 - \prod_c (1 - \rho'_c)$ over the open candidates c of A inside M is the probability of a yes, and e_M is the expected number of answers that a yes costs beyond the one for its FD.

Finding an FD. A yes to $M \to A$ means that an FD $X \to A$ with $X \subseteq M$ is missing from H, and Cert3NF searches for a minimal one by Find (line 15), first with at most $g = \lceil \log_2 |M| \rceil + 1$ tries. A true FD that is not minimal on the data has no candidate of its own, so Cert3NF first tries the left-hand sides inside M that the designer has confirmed for other attributes. A yes to such a try narrows the search to its left-hand side. Cert3NF then guesses the likeliest open candidates inside M, as long as they are likely given $M \to A$ (Appendix B). If the tries miss, it asks about subsets of M by QuickXplain [27], which finds a left-hand side of k attributes with $O(k(1 + \log(|M|/k)))$ further answers. Where most maximal sets asked so far were answered yes, even the yes to $M \to A$ is a waste, since a set whose first guess hits costs two answers and the guess alone costs one. With $\bar{p}$ the rate of maximal sets answered yes so far and h' the rate at which a first guess hits when its set is answered yes, asking the guess first wastes $1 - \bar{p}h'$ answers in expectation and asking the set first wastes $\bar{p}$. Hence, Cert3NF asks the guess first if $\bar{p}(1 + h') > 1$, and the set at once when the guess misses (lines 11–14).

A run ends when no pair needs a question (line 2), that is, when every maximal set of H that holds on r is refuted. By Thm. 4.2, $H^+ = \Sigma^+$, and Cert3NF returns $\mathcal{A}(H)$ with its certificate: the confirmed FDs and the refuted pairs, each refuted by a no answer or by two records of r.

In Ex. 4.3, the statistics make O look like a key and the advisor doubts $N \to D$, so it believes $O \to N$, $O \to C$, $O \to D$ and $N \to C$, with the lowest probability on $O \to C$. Yes answers to $O \to N$ and $N \to C$ settle $O \to C$ before its turn, so the four believed FDs take three answers. One more answer refutes the only maximal

set that holds on the data, NC for D. Since $N_H^+ = NC$, Cert3NF asks about $N \to D$ in place of $NC \to D$, which is the question of the introduction. If the advisor also doubted $O \to D$, the designer would confirm $O \to N$ and $N \to C$, and Cert3NF would ask about $O \to D$ in place of the maximal set ONC for D. The certificate would still take four answers, the price. In both cases, the only no answer falls on the held set NC, and the next result shows that this holds in general.

Proposition 5.1 (Refuted maximal sets). *Let r satisfy Σ and the designer answer truthfully. Every no answer to a maximal set, or to a candidate asked in its place, falls on a held set, and no held set receives two. Hence, Cert3NF receives at most m_r such no answers.*

The no answers to maximal sets thus stay within the m_r no answers of the price. The next result counts all answers.

Proposition 5.2 (Number of answers). *Let r satisfy Σ and the designer answer truthfully, and let $J(k, n) \in O(k(1 + \log(n/k)))$ be the number of answers that QuickXplain needs to find a left-hand side of k attributes among n. The FDs confirmed by a yes to a candidate and those found through a maximal set are different FDs of Σ^+ that Σ_0 does not imply, each with a left-hand side that is minimal under Σ. If there are d such FDs, Cert3NF asks at most*

$$m_r + d + b + \min(m_r, d) + \sum_{X \to A} \big(g + J(|X|, |M|)\big)$$

questions, where $X \to A$ runs over the FDs found by a yes to (M, A).

That is, beyond the m_r no answers to maximal sets and one answer for each of the d FDs, an advisor costs at most b no answers to believed FDs, g wrong tries per FD, and one wrong first guess per refuted set. On our relations, d is only slightly larger than a smallest cover in the median, although it can be exponentially larger in general (Appendix A). The FDs that a yes reveals are found at a logarithmic cost, whatever the advisor says, and no method avoids this cost, since one-bit answers must tell apart all left-hand sides that the data allow.

Proposition 5.3 (Lower bound). *For all $1 \le k < n$, let $R = MA$ with $|M| = n$ and $\Sigma_0 = \emptyset$. There is a relation r over R on which every $\Sigma = \{X \to A\}$ with $X \subseteq M$ and $|X| = k$ holds and has the price $k + 1$, while every method that certifies a design for each of them needs at least $\log_2 \binom{n}{k} \ge k \log_2(n/k)$ answers on one of them.*

Hence, no method certifies every design within a constant factor of the price. Without an advisor, Cert3NF needs at most $1+k+J(k, n)$ answers on these relations. Since every method also pays the price, this is within a constant factor of the best method on these relations. An advisor that believes exactly $X \to A$ brings the cost down to the price.

Partial certificates. If the budget runs out (lines 17–18), Cert3NF certifies what it can. A schema S of $\mathcal{A}(H)$ is certified on its own when every maximal set of the projection $H[S]$ is refuted, which leaves only the preservation of FDs across the schemata open.

Proposition 5.4 (Partial certificate). *Let $H \subseteq \Sigma^+$, let $S \subseteq R$, and let $T \to A \notin \Sigma^+$ for every $A \in S$ and every $T \in \max(H[S], A)$, taken over S. Then Σ^+ and H^+ contain the same FDs over S, so $H[S] = \Sigma[S]$, and if S is a schema of $\mathcal{A}(H)$, it is in 3NF with respect to Σ.*

Without a certificate, Cert3NF returns the sound design $\mathcal{A}(H)$ with its certified schemata. The design is lossless, its certified schemata already declare the true constraints, and a later session continues from its answers.

A designer who errs. The certificate of Thm. 4.2 is a proof, and a wrong answer is a false premise. The *premises* are a non-redundant subset of the yes answers that generates H with Σ_0, and a subset of the no answers that refutes every maximal set of H that holds on r. If all of them are right, so is the certificate. In its *careful mode*, Cert3NF therefore has the designer answer every premise a second time before it certifies, and a third time when the two answers disagree. The net answer of a question is the majority of its answers, and a question with one answer of each kind counts as unanswered. QuickXplain infers the FD it finds from several answers, so the designer is asked about this FD itself before it joins H, which holds net yes answers only. No question is asked more than three times, so a premise is wrong only if two of its answers are. The certificate is issued once no *conflict*, a net no to an FD that H implies, is left and every premise has a majority of at least two answers (Appendix A). The next result bounds the probability of a wrong certificate.

Proposition 5.5 (Wrong certificates). *Let r satisfy Σ, let every answer be wrong independently with probability ε, and let Q be the expected number of different questions that Cert3NF asks in its careful mode. Then Cert3NF issues a wrong certificate with probability at most $Q(3\varepsilon^2 - 2\varepsilon^3)$, which is below $300\varepsilon^2$ within a budget of 100 answers.*

For example, if a designer errs on one answer in fifty, at least seven runs in eight end without a wrong certificate, on any data that satisfy Σ. With an error rate ε_q per question, the bound holds with the mean of $3\varepsilon_q^2 - 2\varepsilon_q^3$ (Appendix A), so a second answer does not help against a designer who repeats every error.

Data that violate true FDs. The certificate also assumes that the data satisfy Σ, while real data violate true FDs in a few records. The data then refute such an FD without a question, so the designer never sees it. We say that $M \to A$ holds on r *up to* θ if at most a share θ of the records has to be removed for $M \to A$ to hold. This share is the measure g_3 of Kivinen and Mannila [29], which discovery of approximate FDs uses [24]. With a *tolerance* θ, Cert3NF asks the designer about every maximal set $M \in \max(H, A)$ for which $M \to A$ holds on r up to θ, and lets the data refute the others. The next result shows that the certificate then stays exact on data that satisfy the true FDs up to θ.

Proposition 5.6 (Tolerance). *Let every FD of Σ^+ hold on r up to θ, and read "holds on r" as "holds on r up to θ" and the candidates as the minimal FDs that hold on r up to θ and that Σ_0 does not imply. Then the data refute no FD of Σ^+, a set of answers that certifies a design proves it good for Σ, and Thm. 4.4(a)–(c) hold.*

Every maximal set that the data no longer refute costs a question, so a tolerance trades questions for robustness.

6 EXPERIMENTS

Our experiments answer the four questions of the introduction, which build one argument. Single errors damage most synthesized designs, so designs must be certified (Q1). No automatic method reliably obtains a good design, so a designer pays the price of a

certificate (Q2). An advisor cuts the answers above this price (Q3), and certificates survive the slips of a designer (Q4).

6.1 Setup

Data. Random schemata and the eight tables of TPC-H [46] measure the damage of single errors (Q1). The experiments use 59 relations with data and documented FDs, with 5 to 25 attributes and 200 to 2.7 million rows (Tab. 8 in Appendix B). Cert3NF was designed on 36 *development* relations, on which its prior is learned, and we evaluate it on the 35 of them that have answers of the LLM and on 23 *confirmatory* relations. The confirmatory relations were chosen from their documentation alone by rules fixed in advance, and never opened during development. Seven of them are benchmarks whose schema an LLM may have memorized (class A), and sixteen are real data with a data dictionary (class B).

Ground truth. The ground truth of a relation consists of the FDs that its publisher documents, as a *full* truth that includes the rules a domain expert has to infer and a *sure* truth of the explicit ones. Together with the declared key as Σ_0, it plays the designer. A design is *good* if it is good for the documented FDs and the declared key (Sec. 4). A certificate is *wrong* if it rests on a false FD or misses a documented FD that holds on the data. Before all runs, a greedy repair removed records until the sure FDs hold (Appendix B). On five confirmatory relations, listed before any run and marked † in the tables, the data still violate a documented FD of the full truth, which then lies outside every certificate.

Methods. We compare Cert3NF with three groups of methods. Its variants isolate the advisor and the order. The variant *with LLM* also reads the answers of the LLM, and the variant *two phases* also reads them but confirms the believed cover first and then asks about the maximal sets, most likely first. The variant *without advisor* asks about the maximal sets alone, in a fixed order, with QuickXplain in place of the guesses. The interactive baselines ask the same designer, but about candidates alone, and also read the answers of the LLM. The *posterior certificate* asks about the candidates of largest marginal under a posterior over closed sets of candidates, and accepts a sound design once it is good for a share $1-\delta$ of the closed sets sampled from the posterior. EC^2 asks about the candidates whose answers separate the most pairs of posterior samples with different designs [18]. To separate our framework from the choice of its questions, the variant EC^2 *order* keeps the candidates, maximal sets and stopping rule of Cert3NF, but chooses the next question by the score of EC^2, without the LLM, and finds the FD behind a yes by QuickXplain alone. The variant *Normalize order* replaces the estimator of Cert3NF by the score of Normalize [42]. The fully automatic methods ask no designer: the mined FDs taken as true (*mined FDs*), the BCNF decomposition along the mined FDs of highest score (*Normalize*), the yes answers of the LLM (*LLM FDs*), a per-FD classifier on the statistics and the answer (*classifier*), the posterior alone, the rules that the LLM states (*rules*), and the designs of the LLM, asked directly (*LLM design*) or as a multi-agent normalizer with a verifier after MARS [30] (*normalizer*). As a reference, *every candidate* lets the designer judge every candidate without a budget.

Protocol. The LLM is Qwen3.5-9B on an NVIDIA RTX 4090 (Appendix B). The candidates are all minimal FDs of the data, mined exactly in the style of HyFD [41]. Cert3NF is deterministic, and every method with a random component runs with three seeds. Five claims, marked (V1) to (V5) where the text states them, are judged per truth and class by rules fixed before any confirmatory result was seen (Appendix B). All hold except one upper quartile of (V4).

Table 1: Damage by a Single Error on Random Schemata with $n \leq 6$ Attributes (in %, $n = 7$ in Parentheses, – No Lossy Design)

	false FD		missing FD	
measure	classical	iCONF	classical	iCONF
over-enforcing	100 (100)	100 (100)	0 (0)	0 (0)
locally under-enforcing	0 (0)	0 (0)	73.1 (64.0)	73.0 (64.0)
lossy	51.1 (62.8)	50.8 (62.6)	0 (0)	0 (0)
not preserving	41.5 (28.0)	41.5 (27.1)	44.5 (59.2)	44.6 (59.2)
not in 3NF	10.4 (10.1)	8.5 (7.8)	30.8 (43.8)	30.7 (43.8)
structural damage	68.6 (73.5)	68.2 (72.8)	68.0 (87.5)	68.1 (87.5)
key loss, of lossy	63.6 (85.5)	64.1 (86.2)	–	–

Table 2: Damage by a Single Accidental FD on TPC-H (in %, iCONF in Parentheses Where It Differs)

table	$\|R\|$	accidental FDs	lossy	not in 3NF	structural damage
region	3	2	100	0	100
nation	4	3	100	0	100
partsupp	5	16	100	0	100
supplier	7	21	85.7	14.3 (4.8)	90.5
customer	8	34	94.1	0	94.1
orders	8	44	100	0	100
part	9	79	86.1	7.6 (3.8)	89.9
lineitem	16	3,670	89.3	0.6 (0.4)	89.8
all	–	3,869	89.5	0.8 (0.5)	90.0

6.2 Q1: Which Errors Must Be Resolved?

We first ask which errors damage a design, and how often.

Random schemata. Tab. 1 shows that one error causes structural damage to about two thirds of the designs for $n \leq 6$, and to up to 87.5% for $n = 7$, whichever algorithm synthesizes them. Indeed, the algorithms differ in their order π alone, and every order walks the same wrong atomic closure with the same wrong keys and only selects among its covers. Hence, robustness to errors has to come from the estimate or from the designer, not from the choice of the synthesis algorithm. The kind of damage depends on the kind of error. As Prop. 3.3 predicts, every false FD over-enforces and half of them make the design lossy, while missing FDs do neither and break preservation or 3NF instead.

TPC-H. Tab. 2 shows that accidental FDs damage the structure even more often than random false FDs: 90.0% of them cause structural damage, against 68.6% on random schemata. Indeed, in every table the primary key K determines each other attribute directly, so $K \to X$ and an accidental FD $X \to a$ derive $K \to a$, which the walk can then remove. Unless a true non-key FD also reaches a or leads from a into X, synthesis then stores a with X instead of with

Table 3: Answers until the Certificate on the Confirmatory Relations, with True Candidates and Price (Thm. 4.4) per Truth; in Parentheses: Answers without a Budget, above 100; – No Certificate (Post. and EC^2: within 100), † a Design That Misses a Documented FD the Data Violate; LLM: Cert3NF with the LLM; EC^2 Ord.: Cert3NF with the Order of Questions of EC^2; Post.: Posterior Certificate; Post. and EC^2 Ask about Candidates Alone, with the LLM; (A): Class A; EC^2 Ord., Post. and EC^2: Median of Three Seeds; $\theta = 0.1$: Cert3NF on the Unrepaired Data with This Tolerance; Underlined: Best; Wavy: Second Best; $\theta = 0.1$ Not Ranked

		full truth									sure truth								
relation	cand.	true	price	Cert3NF	LLM	w/o adv.	EC^2 ord.	post.	EC^2	θ=0.1	true	price	Cert3NF	LLM	w/o adv.	EC^2 ord.	post.	EC^2	θ=0.1
tpcc-orderline (A)	12,130	67	77	(115†)	(142†)	(252†)	(261†)	–	–	(114)	32	70	(120)	95	(168)	(189)	–	–	(121)
tpcc-stock (A)	238	16	33	48	45	59	53	–	–	48	16	33	34	34	59	63	–	–	34
tpcc-customer (A)	272	12	32	41	42	54	69	–	–	41	12	32	41	40	54	63	–	–	41
tweets (A)	10	2	6	7	8	8	10	10	10	7	2	6	7	9	8	10	10	10	7
syn-encounters (A)	635	13	32	51	39	(105)	71	–	–	54	13	32	48	40	(105)	87	–	–	51
syn-claims (A)	201	7	25	39	42	53	60	–	–	45	7	25	37	43	53	57	100	95	43
syn-trans (A)	244	5	27	39	39	57	52	75	90	40	5	27	40	38	57	54	65	70	41
companies	994	0	22	22	22	22	22	–	–	25	0	22	22	22	22	22	–	–	25
fec-cand	718	11	22	35	30	66	33	–	–	43	11	22	29	29	66	47	–	–	36
fec-comm	868	8	20	23	22	62	32	–	–	30	8	20	22	22	62	39	–	–	29
fec-link	657	20	30	41	41	87	65	–	–	55	18	29	37	36	88	54	–	–	50
gleif	4,008	0	24	24	24	24	24	–	–	25	0	24	24	24	24	24	–	–	25
hgnc	761	327	227	(691†)	(723†)	–	–	–	–	(311)	23	23	25	26	27	43	–	–	26
nppes	327	0	8	8	9	8	8	–	–	24	0	8	8	8	8	8	100	–	24
onspd	311	83	53	(194†)	(182†)	(569†)	(356†)	–	–	(157)	20	27	56	54	(125)	(101)	–	–	93
pay-general	741	11	26	36	34	60	74	–	–	70	10	25	34	33	56	64	–	–	46
pay-profile	122	3	18	20	20	30	29	65	65	29	3	18	20	20	30	30	50	50	29
air-airports	557	62	33	72†	69†	(130†)	73†	–	–	100†	62	31	61	63	(115)	61	–	–	63
air-freq	1,574	45	46	(124†)	(118†)	(200†)	(133†)	–	–	(252†)	45	39	49	47	(130)	58	–	–	52
air-navaids	5,271	29	30	39	44	72	63	–	–	46	29	30	38	35	72	49	–	–	42
se-postlinks	1,237	19	43	56	45	(115)	98	–	–	56	19	43	51	46	(115)	(102)	90	95	51
se-posts	1,053	11	35	41	38	100	83	–	100	42	11	35	40	39	100	77	98	93	41
se-users	18	0	8	8	8	8	8	10	10	8	0	8	8	8	8	9	10	10	8
within 100				19/23	19/23	16/23	19/23	4/23	5/23	19/23			22/23	23/23	17/23	20/23	8/23	7/23	22/23
within 50				16/23	18/23	6/23	8/23	2/23	2/23	14/23			19/23	20/23	7/23	10/23	3/23	3/23	17/23

Table 4: Good Designs of the 35 Development and the 23 Confirmatory Relations, by the Median of Three Seeds, under the Full and the Sure Truth in Each Cell; Shaded: at the End of a Budget of 100 Answers; Post.: Posterior Certificate; –: Run on the Confirmatory Relations Only

method	dev.	conf.	method	dev.	conf.
mined FDs	0, 0	0, 0	normalizer	1, 3	3, 2
Normalize	–	5, 4	every candidate	23, 30	18, 23
LLM FDs	0, 0	0, 0	Post.	18, 24	16, 19
classifier	0, 0	2, 4	EC^2	–	16, 17
posterior alone	0, 0	0, 0	EC^2 order	–	18, 22
rules	0, 0	0, 0	Normalize order	–	18, 23
LLM design	0, 0	5, 5	Cert3NF	26, 34	18, 23

K. Since X is not a superkey under Σ, the join on X is lossy, as in Ex. 1.1. The two mechanisms of Thm. 3.4 thus dominate on different data: key loss causes most lossy designs on random schemata (last row of Tab. 1), and path loss causes every lossy design on TPC-H. In summary, a single false or missing FD damages the structure of most synthesized designs, which costs a migration of the data, so that every candidate has to be settled before deployment (Q1).

6.3 Q2: How Many Answers Certify a Design?

Since every candidate must be settled, we first ask whether an automatic method can do so, and then how many answers a designer needs.

Automatic methods. Tab. 4 shows that no fully automatic method reliably obtains a good design. The best of them, the LLM design and Normalize, obtain a good design for at most five of the 23 confirmatory relations, and no other method for more than four. Indeed, a good design has to preserve every true FD and avoid the damage of false ones, and each method fails at least one of the two. The mined FDs, along which Normalize splits, often contain thousands of accidental ones. The LLM refutes up to three quarters of the true candidates of a relation or omits attributes and rules in the designs it writes, and the normalizer infers only part of the rules. Graded by the damage of Sec. 3, the designs of these methods are lossless on at most ten of the 23 confirmatory relations under either truth. Even a designer who judges every candidate obtains a good design only where every true FD is a minimal FD of the data and the data satisfy Σ. These conditions fail on up to a third of the development relations and on the five confirmatory relations marked † in Tab. 3. Hence, the designer is not optional and the candidates are not enough, and the question becomes what a designer has to answer.

Certificates within the budget. Tab. 3 shows that a designer who answers a hundred questions certifies the design of almost every relation: Cert3NF certifies 19 (full) and 22 (sure) of the 23 confirmatory relations. Every certificate is correct where the data

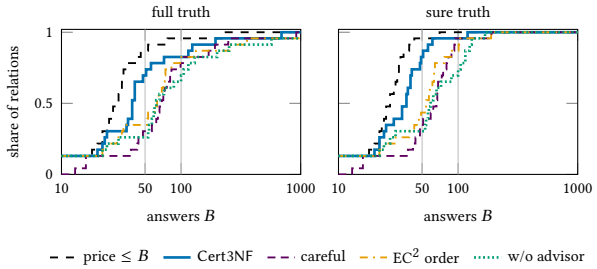


Figure 2: Share of the 23 Confirmatory Relations Certified within B Answers, and Share Whose Price Is at Most B; Grey Lines: 50 and 100 Answers; Careful: the Careful Mode with a Designer Who Never Errs; EC^2 Order: Median of Three Seeds

Table 5: Answers until the Certificate on Synthetic Relations without a Budget (Medians over Three Seeds; True: True Candidates; Share: Share of Entity Attributes among Those outside the Fact Key; Ratio: Answers to Price)

					Cert3NF		w/o advisor
attr.	share	cand.	true	price	answers	ratio	answers
10	0.6	78	5	17	20	1.18	38
20	0.6	16,294	19	42	46	1.11	154
30	0.2	554,220	9	59	64	1.07	113
30	0.4	806,080	20	63	69	1.08	153
30	0.6	145,161	40	61	72	1.18	256
30	0.8	284,217	53	61	73	1.18	268

satisfy the documented FDs, as Thm. 4.2 guarantees (V2), so the answers are what remains to measure. The certified runs stay close to the price of Thm. 4.4, a lower bound for every method that asks yes or no about FDs $Z \rightarrow A$, and the development relations behave alike (Tab. 9).

Interactive baselines. The framework of Cert3NF, more than the choice of questions, makes a certificate affordable. The posterior certificate and EC^2 ask about candidates alone and certify at most 8 of the 23 relations within the budget, while the EC^2 order, with our framework but without the LLM, certifies 19 (full) and 20 (sure) (Tab. 3). Indeed, a no to a maximal set settles every candidate that leads from inside it to an attribute outside it, while the baselines ask about these candidates one by one. The posterior certificate also trusts the candidates and the LLM, and errs where a true FD is not minimal on the data or is refuted by the LLM: it is wrong on 3 (full) and 4 (sure) of the development relations it certifies, while Cert3NF is never wrong (Tab. 9). On the confirmatory relations, no certificate of any method is wrong, so the methods differ in their designs. At the end of the budget, Cert3NF has more good designs than either baseline and certifies all of them under the full truth and all but one under the sure truth (shaded rows of Tab. 4, V5). The baselines certify only about a third of theirs, and nothing tells their other good designs from their bad ones.

The choices of Cert3NF still pay, and most where answers are scarce. Within 50 answers, Cert3NF certifies 16 (full) and 19 (sure) of the 23 relations, against 8 and 10 with the EC^2 order (last row of Tab. 3). Without a budget, it needs a median of 0.70× (full) and 0.65× (sure) the answers of the EC^2 order (V3), and at most half as many answers above the price (Appendix B). Indeed, the EC^2 order starts QuickXplain at once, while Cert3NF first tries a confirmed left-hand side or the likeliest candidate inside the set, which saves a third (full) and about four fifths (sure) of the answers that finding the FD behind a yes costs.

Designs without a certificate. Where the budget runs out before the certificate, the design of Cert3NF is still lossless, since it rests on confirmed FDs alone. In contrast, the posterior certificate and EC^2 rely on unconfirmed FDs where they stop, and a quarter to a third of their uncertified designs are lossy under the full truth. The designs that Cert3NF leaves uncertified are also certified in parts, since most of their schemata are certified on their own (Prop. 5.4).

Without a budget. Fig. 2 shows that only the budget stops the uncertified runs of Cert3NF. With a budget of 1,000 answers instead, Cert3NF certifies every relation (parentheses in Tab. 3 and Tab. 9), almost all within 200 answers, about an hour and a half at the pace of the experts in our study (Sec. 6.5). The certified runs then pay a median of 1.37× (full) and 1.26× (sure) the price on the confirmatory relations, and the upper quartile stays within 2×, except for 2.03× on class B under the full truth (V4). Only hgnc and onspd exceed 3× the price, both under the full truth, where the data violate documented FDs and the designer confirms many FDs with larger left-hand sides in place of each violated one.

Scalability. Tab. 5 shows that the domain and the data, not the number of candidates, set the cost of a certificate. On synthetic snowflake joins whose true FDs hold by construction (Appendix B), the candidates grow from 78 at 10 attributes to about a million on the largest run at 30. Yet the median price is at most 63, and Cert3NF certifies every design correctly at a median of 1.15× the price. Indeed, the accidental candidates multiply with the attributes, but the maximal sets settle them in groups. Hence, the answers follow the price (Spearman's rank correlation 0.79) rather than the candidates (0.49), as fixed before generating the data (Appendix B). Without an advisor, Cert3NF needs up to 3.7× as many answers (last column), so the statistics alone save most of the cost where the names of the attributes carry no meaning. Between two answers, Cert3NF computes for less than a second in the median, also with a million candidates.

Attribute exploration. The closest classical method determines Σ as exactly as a certificate does, but at a multiple of its answers. Its expert confirms an implication or constructs a counterexample [15, 16]. With the smallest counterexamples, exploration does not finish within two hours on eight (full) and nine (sure) confirmatory relations. With the largest, which are maximal sets, each implication asks about a whole conclusion and each counterexample is grown attribute by attribute, which amounts to 7× (full) and 9× (sure) the yes and no answers of Cert3NF without a budget (Appendix B).

Violated FDs. A certificate assumes that the data satisfy Σ, and the tolerance of Prop. 5.6 keeps it exact on data that violate true FDs in a few records, as the columns $\theta = 0.1$ of Tab. 3 show on the data before the repair. Of the five † relations, only air-airports and airfreq keep their †, and every other certificate is correct. On these two, rules that their full truth infers from the documentation fail on 39% and 29% of the records. On the unrepaired data of the three relations

that the repair changed, every certificate without a tolerance misses a documented FD under either truth, while $\theta = 0.01$ already makes them all good. The tolerance also saves the answers that violated FDs cost, as on hgnc, where it needs 0.45× the answers under the full truth. Hence, $\theta = 0.1$, the largest of the three tolerances fixed before these runs, is a good default, at about 4% more answers in the median on data that satisfy the sure truth.

In summary, a certificate costs a designer about 1.3× its price in the median, the price is set by the domain and the data rather than by the number of candidates, and the framework of Cert3NF, more than the choice of its questions, makes the certificate affordable.

6.4 Q3: What Does an Advisor Buy?

Since the certificate does not depend on the advisor, we ask what the advisor contributes and what a wrong one costs.

The statistics. The statistics of the data alone buy most of the savings. Without an advisor, Cert3NF certifies 16 (full) and 17 (sure) of the 23 confirmatory relations within the budget, against 19 and 22 with it (Fig. 2), and without a budget it spends 2.5× and 4.2× as many answers above the price. Indeed, QuickXplain has to find every revealed FD one attribute at a time, while the advisor guesses it in one. Which statistics the advisor starts from matters little: with the untrained score of Normalize instead, Cert3NF certifies 19 (full) and 23 (sure) of the relations within the budget at about as many answers, where Normalize alone obtains five and four good designs (Appendix B).

The order. Choosing the next question after every answer also pays. Even without the LLM, Cert3NF needs a median of 0.89× (full) and 0.85× (sure) the answers of the variant two phases, and fewer answers in total (V1). Indeed, the variant two phases asks first about the maximal sets most likely to hold and pays for the FD behind every yes, while Cert3NF first asks about those that a no is likely to settle, and asks about a likely FD inside a set directly, at one answer instead of two.

The LLM. On top of the statistics, the LLM adds little. With it, Cert3NF certifies one more relation within the budget, under the sure truth, and needs about as many answers without a budget (column LLM of Tab. 3). No certificate is wrong with or without it, so the LLM stays optional. With the same LLM, the EC^2 order gets worse: above the price, it spends 1.6× (full) and 2.2× (sure) as many answers as without it. Indeed, its sampler weighs the samples by all answers of the LLM at once, so that a few samples carry the weight, while Cert3NF asks the LLM only where its answer can change the next question.

A wrong advisor. Where the advisor errs, it costs only what Prop. 5.2 charges it: on all 23 confirmatory relations and under both truths, Cert3NF needs at most b more answers than without an advisor, plus one per maximal set answered yes and per wrong first guess, and certifies at least as many relations (V5). In summary, the statistics buy most of the savings, and a wrong advisor costs a bounded number of answers but never the correctness of a certificate.

Table 6: Certificates within the Budget and Wrong Certificates (✗) of a Designer Who Errs with Probability ε, over Relations and Three Seeds (105 Development and 69 Confirmatory Runs per Cell); Repeated: the Designer Repeats a Wrong Answer; ϕ: Correlation of Two Answers to a Question

		development				confirmatory			
		full		sure		full		sure	
ε	mode	cert.	✗	cert.	✗	cert.	✗	cert.	✗
0.02	trusting	73	31	82	28	43	22	53	26
0.02	careful	47	1	56	0	29	0	42	1
0.05	trusting	50	36	54	39	37	26	42	29
0.05	careful	24	1	25	2	16	0	22	1
0.10	trusting	36	30	36	32	18	14	20	16
0.10	careful	15	6	15	2	10	1	9	1
0.15	trusting	22	19	20	16	15	11	16	12
0.15	careful	13	11	11	8	6	1	7	2
0.05, repeated	careful	25	11	26	12	19	8	26	13
0.02, $\phi = 0.25$	careful	39	4	51	4	35	3	42	5
0.02, $\phi = 0.5$	careful	39	9	47	6	36	7	48	9
0.05, $\phi = 0.25$	careful	25	13	24	10	13	5	17	5
0.05, $\phi = 0.5$	careful	28	15	29	10	17	5	22	9

6.5 Q4: Does a Certificate Survive a Designer Who Errs?

The designer of (Q2) and (Q3) never errs. We now let it answer wrongly with probability $\varepsilon \in \{0.02, 0.05, 0.10, 0.15\}$, independently per answer and with three seeds, and compare the careful mode of Cert3NF with its trusting mode (Alg. 1).

Independent errors. Tab. 6 shows that a certificate that trusts every answer breaks under a few wrong ones, while the careful mode keeps it correct as long as the designer errs rarely. Indeed, a premise is then wrong only if two of its answers are, which happens with probability $3\varepsilon^2 - 2\varepsilon^3$ instead of ε. Over all relations and both truths at $\varepsilon \leq 0.05$, trusting the designer makes more than half of the certificates wrong (237 of 434), while answering every premise twice leaves 6 of 261 wrong (2%), far within the bound of Prop. 5.5. The careful mode certifies fewer runs, but issues more correct certificates, 255 against 197. However, wrong certificates return as ε and the number of answers grow: at $\varepsilon = 0.10$, the careful mode still cuts them from 84% to 20%, but at $\varepsilon = 0.15$, 59% of its certificates are wrong, and without a budget 42% already at $\varepsilon = 0.05$.

Second answers. The careful mode pays for its correctness in answers: with a designer who never errs, it asks every premise once more, at about 1.75× the answers of the trusting mode in the median (Fig. 2), and still certifies 18 (full) and 22 (sure) of the 23 confirmatory relations within the budget.

Correlated errors. Asking again catches only the independent part of the errors, never a designer who repeats a wrong answer (row "repeated" of Tab. 6). Between these two extremes, every question has its own error rate, drawn with mean ε so that two answers to it err together with correlation ϕ (last rows of Tab. 6). At $\varepsilon = 0.02$, wrong certificates grow from 1% at $\phi = 0$ to 18% at $\phi = 0.5$, and at $\varepsilon = 0.05$ they reach 42% already at $\phi = 0.25$, close to a designer who repeats every error.

Table 7: Nine Designers, Three per Condition, on Three Confirmatory Relations (E: Experts; P, U: Practitioners with and without Documentation); Block: Fixed Questions about Candidates and Maximal Sets; $\hat{\phi}$: Correlation of Two Answers to a Question; – Undefined without a Wrong Answer; Answers and Minutes: Means per Session, without Breaks

errors (%)	E	P	U	sessions	E	P	U
block candidates	0	2	2	answers	65	68	75
block maximal sets	0	2	2	minutes	30	45	49
in the sessions	0	10	15	certified, of nine	9	6	4
don't know (%)	0	2	5	wrong, full truth	0	1	1
$\hat{\phi}$	–	0.12	0.18	wrong, sure truth	0	1	1
				good designs	9	7	5

Human designers. Tab. 7 shows that, in a pre-registered study with nine people (Appendix B), experts certify every design, none wrongly, in about half an hour per relation, and that the careful mode catches the slips of practitioners but not their false beliefs. The practitioners erred on 10% of their answers and certified six of nine designs with the documentation, and erred on 15% and certified four without it. The independent designer of Tab. 6 is thus pessimistic: at these error rates and on the same relations, it certifies at most one of nine runs within 100 answers, and wrongly. In the fixed blocks, every condition erred as rarely on maximal sets as on candidates, on at most 2%, so the questions that save the most answers are as easy to answer as the others. Two answers of a practitioner to the same question err together with correlation $\hat{\phi} = 0.12$ and 0.18, far from a designer who repeats every error, so most of their errors are slips. Hence, the careful mode catches these slips, and each wrong certificate of a practitioner rests on an error given twice. No proof from answers can catch such a false belief: a certificate is as correct as the beliefs of its designer.

7 RELATED WORK

Schema design. 3NF synthesis [7, 10] and recent work that optimizes designs for integrity maintenance and update efficiency [33, 51, 55–57] assume that the given FDs are correct. Wang et al. normalize a mediated schema with probabilistic FDs estimated from web sources [49], and Link and Prade scale 3NF and BCNF to degrees of certainty of FDs [32]. Both judge the design against the uncertain input, while we judge it against the unknown true FDs.

Dependency discovery. FD discovery algorithms return all FDs that hold on an instance [24, 40, 41], and Dep-Miner derives them from the agree sets and maximal sets of the data [34]. Wei and Link rank the discovered FDs by the redundancy they cause [50], one of the statistics of our advisor, and Wan and Han discover only the k FDs that cause the most [48]. Others find reliable FDs by statistical tests or models of noise [8, 21, 37, 54], and Normalize scores the discovered FDs to decompose a relation into BCNF [42]. These approaches improve the estimate, and each can serve as the advisor of Cert3NF, as Normalize does in (Q3), but they leave errors that only a designer can settle.

Confirming dependencies. Mannila and Räihä let designers confirm FDs by example from Armstrong relations [6, 38], which help designers recognize the meaningful FDs in a user study of Langeveldt and Link [31] and which De Marchi and Petit build from the records of an existing database [39]. Our questions about maximal sets are thus those of design by example (Sec. 4), but only about the held sets, which the data cannot refute. Formal concept analysis treats FDs as implications over the pairs of records [5], and its attribute exploration, seeded with the agree sets, determines Σ as a certificate does [15, 16]. However, its expert has to construct counterexamples, and its questions follow a minimal basis, not the design (Sec. 6.3). Learning such a theory [2, 14] or a minimal cover of FDs [23] needs, besides membership or entailment queries, equivalence queries, which return counterexamples, or closure queries, which return a whole closed set [3]. In contrast, our designer only answers yes or no, and a shortest certificate is a smallest teaching set for the true FDs given the data (Sec. 4). Guided data repair [52] and UGuide [45] also ask users about repairs and FDs within a budget, but to find the errors in the data that the FDs detect.

LLMs as oracle or designer. LLMs predict data correlations from column names [47], and serve as the oracle that decides which statistically strong FDs are meaningful [53], generates ontological FDs [9] or detects foreign keys [13, 44]. As designers, they normalize schemata [26], where inferring the FDs is the main bottleneck [30], or draft domain models, also with a human who answers questions [11, 43]. In both roles nothing bounds the risk of a wrong result, and in our experiments neither obtains a good design for more than five of the 23 confirmatory relations.

In summary, earlier work either designs from estimated FDs or confirms FDs without regard to the design, and none certifies a design against the true FDs. Our work is the first to do so, to price this certificate, and to let the data and an unreliable advisor lower its cost but not its correctness.

8 CONCLUSION AND FUTURE WORK

Starting from the observation that estimated FD sets are wrong, we have shown how a designer certifies a schema design against the true FDs at a price set by the domain and the data rather than by the candidates. A confirmed cover and refuted maximal sets settle the candidates in groups, and a smallest cover with the held sets is a shortest proof from yes and no answers. When the data form an Armstrong relation for the true FDs, the certificate is a confirmed smallest cover, the input of classical design. Asking every premise twice catches a designer's slips, so that a certificate is as correct as the beliefs of its designer.

Future work will address i) designers who answer by example rather than by yes or no, as with Armstrong relations [38], ii) the certification of several relations linked by foreign keys and of relations too wide to mine every candidate, iii) certificates for BCNF, and iv) stronger models as advisors. Our results show that the answers of a designer who knows the domain follow the length of the proof, and that advice that may be wrong can only change the order of the questions.

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A PROOFS

We prove the results in the order of the paper. Losslessness is decided by the chase of the tableau of $\mathcal{D}$ [1, 36], which has one row per schema S, with the distinguished symbol of column A for every $A \in S$ and fresh symbols elsewhere. The chase with Σ equates the symbols in column A of rows that agree on X, for $X \to A \in \Sigma$, and prefers the distinguished symbol. The design $\mathcal{D}$ is lossless if and only if some row ends with distinguished symbols only. Several proofs reuse the argument of Lem. 2.1. It chases the row of the schema S that contains a key along a derivation of $S^+_\Sigma = R$ by the FDs of F. When the derivation applies $X \to A$, embedded in S', the rows of S and S' agree on X in distinguished symbols, so the row of S receives the distinguished symbol in column A.

Proof of Lem. 2.3. The first claim holds because Σ_a, the keys and the fixed choices depend only on Σ^+. The union of the output is R: an attribute in no kept schema occurs in no FD of G and so lies in every key, hence no kept schema contains a key, and the added key contains the attribute. The output also embeds G and contains a key, so it is lossless and dependency-preserving by Lem. 2.1. Each kept schema XA is in 3NF: X is a key of XA by the minimality of X, and a non-trivial FD $Z \to B$ within XA either has $B = A$, so that $Z \subseteq X$ and hence $Z = X$, or has $B \in X$, which is prime. The key schema K carries no non-trivial FD, since $K \setminus \{B\}$ would otherwise be a superkey of R. For the last claim, the kept covers are non-redundant subsets of Σ_a by construction. Conversely, a non-redundant cover $G_0 \subseteq \Sigma_a$ is kept by an order that walks $\Sigma_a \setminus G_0$ first: each of these FDs is implied by $G_0 \subseteq G$ and removed, after which $G = G_0$ and no FD of G_0 is redundant. □

Proof of Prop. 3.3. (a) Every relation that satisfies Σ satisfies $\hat{\Sigma}$ and is therefore the join of its projections.

(b) If some declared FD is not in Σ^+, it is in $\hat{\Sigma}^+$, so $\hat{\Sigma} \not\subseteq \Sigma^+$. Conversely, if every declared FD is in Σ^+, preservation gives $\hat{\Sigma} \subseteq (\bigcup_S \hat{\Sigma}[S])^+ \subseteq \Sigma^+$. For the rejection, let $Y \to B \in \hat{\Sigma}[S] \setminus \Sigma^+$ and take two records that agree exactly on Y^+_Σ. They satisfy Σ, since the set on which they agree is closed, but they violate $Y \to B$, and so does their projection onto $S \supseteq YB$.

(c) Let $\sigma \in \Sigma^+ \setminus \hat{\Sigma}^+$ and let r be a two-tuple relation that satisfies $\hat{\Sigma}$ and violates σ. The projections of r satisfy the declared constraints. If they were the projections of some s that satisfies Σ, then s would satisfy $\hat{\Sigma}$, and losslessness with respect to $\hat{\Sigma}$ would give $s \bowtie_S s[S] \bowtie_S r[S] = r$, which violates σ. □

Proof of Prop. 3.6. Since $\mathcal{A}(\Sigma')$ and $\mathcal{A}(\Sigma'')$ preserve their inputs by Lem. 2.3, we have

$$\Sigma' \subseteq (\textstyle\bigcup_S \Sigma'[S])^+ = (\bigcup_S \Sigma''[S])^+ \subseteq \Sigma''^+,$$

and $\Sigma'' \subseteq \Sigma'^+$ holds symmetrically. □

The proof of Thm. 3.4 rests on two lemmas, which describe what one more FD does to closures and to the chase.

Lemma A.1 (One more FD). *Let $\hat{\Sigma} = \Sigma \cup \{\sigma\}$. Then $Z^+_{\hat{\Sigma}} = Z^+_\Sigma$ if $X \not\subseteq Z^+_\Sigma$, and $Z^+_{\hat{\Sigma}} = (Za)^+_\Sigma$ otherwise.*

Proof. Let Z' denote the right-hand side, which is contained in $Z^+_{\hat{\Sigma}}$ since $\hat{\Sigma} \supseteq \Sigma$ and $a \in Z^+_{\hat{\Sigma}}$ whenever $X \subseteq Z^+_\Sigma$. Conversely, Z' contains Z, is closed under Σ, and is closed under σ, because $X \not\subseteq Z'$ in the first case and $a \in Z'$ in the second, so $Z^+_{\hat{\Sigma}} \subseteq Z'$. □

Let $\hat{\Sigma}$ and $\hat{\mathcal{D}}$ be as in Thm. 3.4, let $\hat{G}$ be the cover that $\mathcal{A}_\pi$ keeps on $\hat{\Sigma}$, and let W_S be the set of columns in which the row of schema S holds distinguished symbols after the chase of the tableau of $\hat{\mathcal{D}}$ with Σ.

Lemma A.2 (Rows and superkeys). *(a) If $X \not\subseteq S^+_\Sigma$ for a schema $S \in \hat{\mathcal{D}}$, then $W_S = S^+_\Sigma = S^+_{\hat{\Sigma}}$.*

(b) If some schema of $\hat{\mathcal{D}}$ is a superkey with respect to $\Sigma[\hat{\mathcal{D}}]$, then $\hat{\mathcal{D}}$ is lossless with respect to Σ. If $\hat{\mathcal{D}}$ is lossless with respect to Σ, then some schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ. If $\hat{\mathcal{D}}$ preserves Σ, then $\Sigma[\hat{\mathcal{D}}]$ and Σ have the same superkeys, and the three conditions are equivalent.

Proof. (a) For the upper bound $W_S \subseteq S^+_\Sigma$, let t and s be two records that agree exactly on S^+_Σ, so that $\{t, s\}$ satisfies Σ. Mapping the row of S to s, every other row to t, and each distinguished symbol to the value of t in its column is a homomorphism from the tableau into $\{t, s\}$, because s and t agree on S. Every chase step equates two symbols that this homomorphism already maps to the same value, so the chased tableau maps into $\{t, s\}$ as well. A column in which the row of S is distinguished is therefore one in which s and t agree, that is, a column in S^+_Σ. For the lower bound, $S^+_{\hat{\Sigma}} = S^+_\Sigma$ by Lem. A.1, and $S^+_{\hat{\Sigma}}$ is derived from S by the FDs of $\hat{G}$. Every FD $Z \to C$ used in this derivation has $Z \subseteq S^+_\Sigma$, so $X \not\subseteq Z^+_\Sigma$, and it is in Σ^+ by Lem. A.1, and as an FD of $\hat{G}$ it is embedded in some schema of $\hat{\mathcal{D}}$. Following the derivation as in the proof of Lem. 2.1, the row of S receives the distinguished symbol in every column of $S^+_{\hat{\Sigma}}$.

(b) The first statement follows from the argument of Lem. 2.1, since every FD of $\Sigma[\hat{\mathcal{D}}]$ is true and embedded in a schema of $\hat{\mathcal{D}}$. For the second, a row of distinguished symbols only belongs to a schema S with $R = W_S \subseteq S^+_\Sigma$ by the upper bound in (a). If $\hat{\mathcal{D}}$ preserves Σ, then $(\Sigma[\hat{\mathcal{D}}])^+ = \Sigma^+$, and the three conditions are equivalent. □

Proof of Thm. 3.4. (a) Let $S^+_{\hat{\Sigma}} = R$. If $a \in S^+_{\hat{\Sigma}}$, then $S^+_\Sigma = S^+_{\hat{\Sigma}} = R$ by Lem. A.1, whether or not $X \subseteq S^+_\Sigma$, and conversely $S^+_\Sigma = R$ contains a. A schema with $S^+_{\hat{\Sigma}} \neq R$ is not a superkey under Σ either, since $S^+_\Sigma \subseteq S^+_{\hat{\Sigma}}$. This proves the equivalence, and lossiness follows from Lem. A.2(b). Finally, $\mathcal{A}_\pi$ ensures that some schema contains a minimal key K of $(R, \hat{\Sigma})$, and under key loss $K^+_\Sigma \neq R$, since that schema would otherwise be a superkey under Σ.

(b) The first statement is the contrapositive of the last sentence of Lem. A.2(b). For the second, suppose that $Y \to B \in \Sigma^+$ is not implied by $\Sigma[\hat{\mathcal{D}}]$ while $X \not\subseteq Y^+_\Sigma$. Then $Y^+_{\hat{\Sigma}} = Y^+_\Sigma$ by Lem. A.1, and as in the proof of Lem. A.2(a) the derivation of B from Y by $\hat{G}$ only uses FDs of Σ^+ that are embedded in schemata S' of $\hat{\mathcal{D}}$. Each of them is implied by $\Sigma[S']$, so $\Sigma[\hat{\mathcal{D}}]$ implies $Y \to B$, a contradiction. □

Proof of Thm. 4.2. Suppose that $M \to A \notin \Sigma^+$ for every A and every $M \in \max(H, A)$, and let $X \to A \in \Sigma^+$. If H did not imply $X \to A$, then $X \subseteq M$ for some $M \in \max(H, A)$, and $M \to A$ would follow from $X \to A$ by augmentation, a contradiction. Hence, $\Sigma^+ \subseteq H^+$, and $H \subseteq \Sigma^+$ gives equality. Conversely, if $H^+ = \Sigma^+$ and

$M \in \max(H, A)$, then $M \to A \notin H^+ = \Sigma^+$. Finally, validity depends only on the closure of the FD set by its definition, and so does $\mathcal{A}$ by Lem. 2.3, so a design that is good for H is good for Σ. The atomic closure also depends only on the closure, so $H[S] = \Sigma[S]$ for every schema S. □

Proof of Thm. 4.4. (a) If M contains the left-hand side of a candidate $X \to A$, then $M \to A$ holds on r by augmentation, since every candidate holds on r. Conversely, if $M \to A$ holds on r, let $X \subseteq M$ be minimal with $X \to A$ holding on r. Since $H \supseteq \Sigma_0$ does not imply $M \to A$, the FD $X \to A$ is not implied by Σ_0, so it is a minimal FD of r outside ${\Sigma_0}^+$, that is, a candidate.

(b) After the yes answers, $H = \Sigma_0 \cup F$, so $H^+ = \Sigma^+$ and $\max(H, A) = \max(\Sigma, A)$ for every A. A maximal set $M \in \max(\Sigma, A)$ for which $M \to A$ fails on r is refuted by two records of r, which satisfies Σ, so $M \to A \notin \Sigma^+$ without a question, and the remaining ones are the held sets. Let M be held, let $M \to A$ be its no answer, and let $M \in \max(\Sigma, A')$ with $M \to A'$ holding on r. Then $M = M_H^+$, since M is closed under Σ, and $A \in (MA')_H^+$, since $(MA')_\Sigma^+$ is a closed set that contains M and $A' \notin M$, so it contains A by the maximality of M for A. Hence, the no answer to $M \to A$ refutes $M \to A'$, every maximal set of H is false, and Thm. 4.2 certifies every design that is good for H, that is, for Σ.

(c) Consider a set of answers that certifies a design. Every maximal set of H is false, so $(\Sigma_0 \cup \Sigma_y)^+ = H^+ = \Sigma^+$ by Thm. 4.2. The held sets are then the $M \in \max(H, A)$ for which $M \to A$ holds on r, each refuted by a no answer to some $Z \to B$ with $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$. If $A \in Z_H^+$, then $B \in (ZA)_H^+ = Z_H^+$ and $Z \to B \in H^+ = \Sigma^+$, which contradicts the truthful no. Hence, Z_H^+ is a closed set that contains M and not A, and $M = Z_H^+$ by the maximality of M. Thus a no answer to $Z \to B$ refutes only pairs (M, A) with $M = Z_H^+$, that is, pairs of one held set, and a certificate has at least one no answer per held set.

(d) Let a set of answers certify a design that is good for Σ, so that $H^+ = \Sigma^+$ as in (c), and let Σ' be a closed set of FDs that holds on r, contains H, and contains no FD answered no. Then $\Sigma^+ \subseteq \Sigma'$. If $X \to A \in \Sigma' \setminus \Sigma^+$, then $X \subseteq M$ for some $M \in \max(\Sigma, A) = \max(H, A)$, and $M \to A \in \Sigma'$ holds on r, so a no answer to some $Z \to B$ refutes it. That is, $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$, so $Z \to B \in \Sigma'$, a contradiction. Hence, $\Sigma' = \Sigma^+$. Conversely, let Σ^+ be the only such set. The closure of H is one, since r satisfies Σ and $H^+ \subseteq \Sigma^+$ contains no FD answered no, so $H^+ = \Sigma^+$. For $M \in \max(H, A)$ with $M \to A$ holding on r, the closure of $H \cup \{M \to A\}$ holds on r and differs from Σ^+, so it contains an FD $Z \to B$ answered no. Since $B \notin Z_H^+$, this requires $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$, so the no answer refutes $M \to A$. Hence, every maximal set of H that holds on r is refuted, and the answers certify every design that is good for H, that is, for Σ. □

Proof of Prop. 5.1. A no answer to a candidate $X \to A$ with $X_H^+ = M$ is a no answer to $M \to A$, since $H \subseteq \Sigma^+$ and $M = X_H^+$, so that $M \to A$ would imply $X \to A$. Let Cert3NF ask about $M \to A$ with $M \in \max(H, A)$ and receive no. Since $H \subseteq \Sigma^+$ and $M \to A \notin \Sigma^+$, $M \subseteq M'$ for some $M' \in \max(\Sigma, A)$; M' is closed under Σ, hence under H, and does not contain A, so $M = M'$ by the maximality of M in $\max(H, A)$. Cert3NF asks only about maximal sets that contain the left-hand side of a candidate, so $M \to A$ holds on r by Thm. 4.4(a), and M is held. For every attribute $A' \notin M$, $(MA')_H^+$ is a closed set that strictly contains M, so it contains A by the maximality of M, and it only grows with H. Hence, the no answer to $M \to A$ refutes $M \to A'$ under every later H, so no later question concerns M. □

Proof of Prop. 5.2. Every candidate holds on r and is minimal there, so a true candidate has a minimal left-hand side under Σ as well, since a smaller true left-hand side would hold on r. A yes answer to a candidate therefore puts into H a true candidate with a minimal left-hand side that H did not imply, and QuickXplain returns a minimal $X \subseteq M$ with $X \to A \in \Sigma^+$, which H did not imply at the yes to $M \to A$, since M is closed under H and does not contain A. Once $X \to A$ is in H, no later maximal set for A contains X, so the same FD is not found twice, and no later question concerns a candidate in H; hence the d FDs are different, and Σ_0 implies none of them. The questions are of five kinds. Believed FDs receive at most b no answers and one yes per FD. Maximal sets and the candidates asked in their place receive at most m_r no answers by Prop. 5.1, and one yes per FD. After a yes to $M \to A$, there are at most g tries, including confirmed left-hand sides, whose yes narrows the search to them, and $J(|X|, |M|)$ questions of QuickXplain if the tries miss. A first guess before a maximal set receives one yes per FD, and a miss counts among the g tries if the set holds. If the set is refuted, the miss is charged to that set, which gives at most m_r such misses. They are also at most d: Cert3NF asks the guess first only if $\bar{p}(1 + h') > 1$ (Sec. 5), and since $h' < 1$, this requires $\bar{p} > 1/2$, that is, more maximal sets answered yes than refuted ones so far, as $\bar{p}$ counts one pseudo-answer of each kind, so the t-th such miss follows at least t sets answered yes, each with an FD of its own. Summing gives the bound. The count d can be exponential in the size of a smallest cover: if Σ consists of $A_i \to B_i$ and $B_i \to A_i$ for $i = 1, \ldots, n$ and of $A_1 \cdots A_n \to C$, a run that settles C before the equivalences confirms one FD for each of the 2^n ways to pick A_i or B_i. □

Proof of Prop. 5.3. Let Σ' consist of the FDs $Y \to A$ with $Y \subseteq M$ and $|Y| = k$, and let r be an Armstrong relation for Σ' [6], on which an FD holds exactly when Σ' implies it. For a k-subset X of M, $\Sigma = \{X \to A\}$ holds on r, and a smallest F has one FD. The maximal sets of Σ are $M \setminus \{x\}$ for A with $x \in X$, and $R \setminus \{B\}$ for $B \in M$. Since $n - 1 \geq k$, Σ' implies $M \setminus \{x\} \to A$, while Σ' implies no FD $R \setminus \{B\} \to B$. Hence, the held sets are the k sets $M \setminus \{x\}$, and the price is $k + 1$. A method chooses each question from the earlier answers, so its runs form a binary tree, and two sets $X \neq X'$ whose runs end in the same leaf receive the same answers. These answers are truthful for both $\{X \to A\}$ and $\{X' \to A\}$, which hold on r and have different closures, so by Thm. 4.4(d) they certify a design for at most one of them. Hence, the tree has at least $\binom{n}{k}$ leaves and a depth of at least $\log_2 \binom{n}{k} \geq k \log_2(n/k)$. □

Proof of Prop. 5.4. For $T \subseteq S$, $T_{H[S]}^+ = T_H^+ \cap S$, since $H[S]$ is a cover of the FDs of H^+ within S. Hence, every $T \subseteq S$ with $A \notin T_H^+$ lies within a member of $\max(H[S], A)$. Let $X \to A \in \Sigma^+$ with $XA \subseteq S$. If H did not imply it, then $X \subseteq T$ for some $T \in \max(H[S], A)$, and $T \to A$ would follow from $X \to A$ by augmentation, a contradiction. With $H \subseteq \Sigma^+$, the two closures contain the same FDs over S. Whether S is in 3NF depends on the

Table 8: The Relations with Data: Attributes, Rows, Candidates, and True Candidates under the Full/Sure Truth; (A) Marks Class A

development										confirmatory				
relation	$\|R\|$	rows	cand.	true	relation	$\|R\|$	rows	cand.	true	relation	$\|R\|$	rows	cand.	true
food	17	312,847	78	31/7	sec	25	999,991	428	16/15	tpcc-orderline (A)	24	300,040	12,130	67/32
hospital	16	1,000	145	54/22	scorecard	25	160,030	5,169	64/33	tpcc-stock (A)	17	100,000	238	16/16
pricepaid	15	926,485	34	13/4	epa	23	381,023	1,895	25/13	tpcc-customer (A)	14	30,000	272	12/12
ssb-customer	8	3,000	24	12/10	pums	25	393,725	2,056	820/8	tweets (A)	5	20,000	10	2/2
ssb-date	17	2,557	209	205/113	chicago	22	259,629	314	21/13	syn-encounters (A)	24	80,771	635	13/13
ssb-lineorder	16	600,572	3,814	13/5	usda	21	644,125	161	69/25	syn-claims (A)	23	153,806	201	7/7
ssb-part	9	20,000	86	3/3	auctionmark	22	34,743	4,243	36/15	syn-trans (A)	22	1,000,000	244	5/5
ssb-supplier	7	200	21	11/9	seats	24	77,399	4,640	292/19	companies	25	849,999	994	0/0
tpcds-core	12	2,685,687	48	23/13	tatp	18	376,152	909	28/28	fec-cand	20	9,796	718	11/11
tpch-lineitem	23	600,572	36,883	10/10	simplewiki	17	999,656	139	36/30	fec-comm	20	20,937	868	8/8
tpch-orders	20	150,000	484	44/44	chembl	25	999,405	385	204/43	fec-link	24	8,619	657	20/18
tpch-partsupp	19	80,000	2,405	25/25	musicbrainz	25	1,000,000	85	71/61	gleif	25	1,000,000	4,008	0/0
tpcds-customer	25	100,000	68,156	44/44	pluto	25	858,284	1,572	202/46	hgnc	24	45,289	761	327/23
ldbc-comment	25	1,000,000	1,784	26/26	fhrs	22	579,855	1,488	25/25	nppes	24	1,000,000	327	0/0
ldbc-study	24	7,949	1,249	39/39	geonames	25	1,000,000	2,900	12/12	onspd	25	998,456	311	83/20
gtfs	24	1,000,000	1,170	23/23	eia-860	25	27,919	1,097	14/14	pay-general	25	999,988	741	11/10
faa	25	317,999	695	14/14	stats19	24	128,272	1,670	42/42	pay-profile	25	1,000,000	122	3/3
abs	17	368,286	78	59/59	ghcn-daily	19	999,369	351	13/13	air-airports	24	86,335	557	62/62
										air-freq	24	30,390	1,574	45/45
										air-navaids	25	10,351	5,271	29/29
										se-postlinks	24	1,581	1,237	19/19
										se-posts	25	15,283	1,053	11/11
										se-users	8	33,730	18	0/0

FDs over S alone, so S is in 3NF with respect to Σ as it is with respect to H by Lem. 2.3, and $\Sigma[S] = H[S]$, since both consist of the FDs over S that are minimal in the common closure. □

The careful mode of Sec. 5 settles conflicting answers as follows. A net no to an FD that H implies is a *conflict*. It is settled by asking again the question whose majority is smallest, among the no and a minimal set of yes answers that imply the FD. A changed net answer rebuilds H, the refutations and the maximal sets, and the loop continues. The certificate is issued when no pair needs a question, no question is unanswered, no conflict is left, and every premise has a majority of at least two answers. A conflict whose questions have all been asked three times ends the run without a certificate.

Proof of Prop. 5.5. If every premise is right, then $H^+ \subseteq \Sigma^+$, since H is generated by Σ_0 and the yes premises, and every maximal set of H is false, since the no premises refute those that hold on r and r, which satisfies Σ, refutes the others; hence $H^+ = \Sigma^+$ by Thm. 4.2, and the design is good for Σ. The certificate is therefore wrong only if some premise is wrong. Think of the answers as drawn in advance, three for every question, each wrong independently with probability ε, of which Cert3NF reads the first one, two or three, since no question receives more than three answers. A wrong premise is a wrong majority of at least two answers, which requires at least two of the three drawn answers of its question to be wrong, with probability $3\varepsilon^2(1-\varepsilon) + \varepsilon^3 = 3\varepsilon^2 - 2\varepsilon^3$. Number the different questions in the order in which they are first asked. Whether a j-th question is asked, and which one, depends only on the answers read before its first answer, which are independent of its own drawn answers. Hence, the probability that a j-th question is asked and two of its drawn answers are wrong is $\Pr[Q_r \geq j](3\varepsilon^2 - 2\varepsilon^3)$, where Q_r is the number of different questions of the run, and the sum over j is $Q(3\varepsilon^2 - 2\varepsilon^3)$. Within a budget of 100 answers, $Q \leq 100$, and $100(3\varepsilon^2 - 2\varepsilon^3) < 300\varepsilon^2$. With an error rate ε_q per question, drawn independently of the other questions, the same argument applies with the probability $3\mathrm{E}[\varepsilon_q^2] - 2\mathrm{E}[\varepsilon_q^3]$ that two of the three drawn answers are wrong. □

Proof of Prop. 5.6. The data refute an FD only if it fails on r up to θ, so the FD is not in Σ^+. For $X \subseteq M$, every group of records that agree on M lies within a group that agrees on X, so the records that remain after the removal for $X \to A$ also satisfy $M \to A$. Hence, $M \to A$ holds up to θ whenever $X \to A$ does, and (a) follows as in the proof of Thm. 4.4 with a minimal $X \subseteq M$ for which $X \to A$ holds up to θ. A set of answers that certifies a design refutes every maximal set of H that holds on r up to θ, and the other maximal sets are refuted by the data and hence false, so the design is good for Σ by Thm. 4.2, which does not refer to the data. The proofs of (b) and (c) use the data only in this way, to refute the maximal sets whose FDs fail on r, and argue about closures otherwise, so they carry over unchanged. Part (d) does not, since the FDs that hold on r up to θ need not be closed under implication. □

B EXPERIMENTAL DETAILS

This appendix gives the protocol of Sec. 6. The artifact holds the implementation, the mined relations with their ground truth and every raw result.

Environment. Synthesis, Cert3NF and the estimators were implemented in Python. The LLM is Qwen3.5-9B in FP8, served by vLLM 0.31.0 on one NVIDIA RTX 4090 with 24 GB in each of three servers, and every other computation ran on the CPUs of these servers, an AMD Ryzen 7 9700X or 7700X with eight cores and 16 threads, with 60 GB of memory, under Ubuntu 26.04. Times are wall-clock per run and include the advisor, but not the mining of the candidates and the answers of the model, which are computed once beforehand. The second LLM is gpt-oss-20b in its released

MXFP4 weights with medium reasoning effort, served the same way and asked the same prompts, and the error rates of its advisor are estimated from its answers on the development relations, as for Qwen. The model answers one prompt per candidate, which names the attributes, states Σ_0 and shows a few rows, with yes or no and a confidence. Every candidate of every relation is answered once, with a fixed seed per prompt, and Cert3NF reads the answer of a candidate only when it queries the model. The normalizer runs an evidence, a diagnosis and a generator agent of the same model with up to 32,768 output tokens per call, and a deterministic verifier that checks the design against the inferred FDs and returns it for at most two repairs.

Random schemata and TPC-H. For the random schemata we drew Σ over n attributes with 2 to $n + 1$ FDs, whose left-hand sides have one to three attributes, 300 times for each $n \in \{4, 5, 6\}$ and 150 times for $n = 7$. Into each we injected, one at a time, up to 12 random false FDs $X \to A \notin \Sigma^+$ with $|X| \leq 2$ and every missing FD of the canonical cover. This gave 10,737 false and 2,958 missing FDs for $n \leq 6$, and 1,800 and 655 for $n = 7$. On TPC-H, the true FDs follow from the specification: besides the primary keys, the names of customers, suppliers, nations and regions are keys, the brand determines the manufacturer, the country code of a phone number determines the nation, the part together with the quantity or the extended price determines the other, and the ship date determines the line status. HyFD discovers 3,945 minimal FDs on the eight tables, of which 76 follow from the specification, and we injected each of the 3,869 accidental ones alone.

Development relations. The first block of development relations in Tab. 8 consists of three denormalized joins that cover the tables of TPC-H, the five tables of SSB, a core of TPC-DS, and the real data sets food (Chicago food inspections), hospital and pricepaid (UK land registry). Their ground truth is given by the specification of the benchmark or by the rules that the publisher documents. The relations of the other two blocks were chosen and prepared by the rules below: a customer cluster of TPC-DS, the comments and the study relation of LDBC SNB, the GTFS feed of Auckland Transport, the FAA aircraft registry, the ABS statistical geography, SEC financial statement data sets, the College Scorecard, EPA air quality, the ACS PUMS of California, Chicago crimes and the USDA nutrient database, and the items, flights and subscribers of the AuctionMark, SEATS and TATP benchmarks, the category links of the Simple English Wikipedia, the activities of ChEMBL, the recordings of MusicBrainz, the tax lots of NYC PLUTO, the establishments of the UK Food Hygiene Rating Scheme (fhrs), the places of GeoNames, the generators of EIA-860, the casualties of STATS19 and the daily observations of GHCN-Daily. For the results on the first 24 development relations, the learned components of Cert3NF, the prior and the error rates of the model, were trained on the first twelve, and for the last twelve on the first 24, in both cases leaving out the relation under test. For the confirmatory relations they are trained once on all 36, and the forms of the prior and of the error model of the LLM were chosen by their held-out log-loss before any confirmatory answer was read. The model has no answers for tpcds-customer, whose 68,156 candidates were not all answered, so this relation only trains the prior.

The advisor and the search. The estimator is a logistic regression on six statistics of a candidate $X \to A$: the size of X, the share of records that $X \to A$ makes redundant [50], whether X is unique on r, the logarithms of the numbers of distinct values of X and of A, each relative to the logarithm of the number of records, and the difference of the two. It is trained like the other learned components. The advisor recalibrates the estimator as $\mathrm{logit}^{-1}(\kappa \cdot \mathrm{logit}\, \rho_c + \lambda)$, with a prior on (κ, λ) learned on other schemata and updated by the designer's answers. A candidate $X \to A$ can hold only if X is a determinant. A yes on a candidate of X raises the other candidates of X to at least η, the rate at which a candidate holds given that another one of X holds, learned on other schemata and updated during the run, and a no lowers them by Bayes' rule.

In Alg. 1, Ask costs no answer when H, an earlier no or the data decide the question, and otherwise puts a yes into H and lets a no refute every FD that it refutes (Sec. 4). In the search Find for an FD inside M, Cert3NF tries confirmed left-hand sides while such tries keep hitting in the run, and guesses the likeliest open candidate inside M while its probability given $M \to A$ is at least $\max(0.2, 1/g)$. QuickXplain orders the attributes of M by how much probability the open candidates place on them. The rates $\bar{p}$ and h' of Sec. 5 count one pseudo-answer of each kind. Of the parts of the order, modeling whether each left-hand side is a determinant saves most, 11% (full) and 18% (sure) of the answers on the confirmatory relations, and asking the first guess before its set and trying confirmed left-hand sides save 7% each under the full truth. After every FD found, the maximal sets of H are recomputed from the minimal keys of $H \cup \{A \to B : B \in R\}$ [35] by transversals. In the careful mode, every iteration first asks again a question whose answers tie or that takes part in a conflict, Find asks about the FD that QuickXplain returns before it joins H, and once no pair is left, every premise with a single answer is asked again before the certificate is issued.

Confirmatory relations. The confirmatory relations were chosen from their documentation alone, before any data were opened, by rules fixed in advance: public, with a specification or data dictionary that states keys, code tables and hierarchies, outside the development relations and the collections used in related work, at least six benchmarks with a published schema (class A) and the rest real data (class B), at most three relations per publisher, each with a root table of its own. Each relation was specified before its data were opened: which tables, which documented foreign keys to join along, which columns to keep, at most 25 columns, at least a thousand rows, and a sample of one million rows with a fixed seed where there were more. Class A consists of the order lines, stock and customers of TPC-C and the tweets of the Twitter benchmark, generated by BenchBase at scale factor one, and the encounters, claims and claim transactions of Synthea. Class B consists of the companies of Companies House, the candidates, committees and linkages of the FEC, the legal entities of GLEIF, the genes of HGNC, the providers of NPPES, the postcodes of ONSPD, the general payments and recipient profiles of CMS Open Payments, the airports, frequencies and navaids of OurAirports, and the post links, posts and users of a Stack Exchange site. The ground truth is the projection of the documented FDs onto the chosen columns, each with a quotation from the documentation, written before the data were opened and checked by domain experts. An FD that the documentation states explicitly is sure, and one that has to be inferred is not. The data were cleaned as the development relations were, and a greedy repair removed records until the sure FDs hold. It changed

only onspd, pay-general and air-navaids, removing 1,544, 12 and 657 records, at most 6% of a relation. On the unrepaired data of these three relations, the documented FDs fail on at most 0.15% of the records, except one rule of air-navaids on 6%. Sec. 6.3 also reports Cert3NF on the unrepaired data, with and without the tolerances $\theta \in \{0.001, 0.01, 0.1\}$, which were fixed before any run with $\theta > 0$. After mining, a sample of at most 200 candidates per relation, 4,139 in all, was audited blindly against the documentation, without access to the ground truth or to any answer of the model. The audit refuted six documented FDs and added nine candidates that the documentation implies, and inferred FDs were promoted to sure on two relations. Before any method ran, we listed the documented FDs that the data violate, 28 inferred FDs on tpcc-orderline, hgnc, onspd, air-airports and air-freq, and checked that every documented FD that holds on the data is minimal there.

Simulated designer. The documented FDs and the declared key play the designer: a question whether $Z \to A$ holds is answered yes if and only if they imply it, and a question that the data refute is not asked. For (Q4), the simulated designer answers wrongly with probability ε, independently on every answer. Whether an answer is wrong is decided by a hash of the relation, the seed, the question, ε and the number of times the question was asked, so that every method meets the same errors. A designer who *repeats* an error gives the first wrong answer again when asked again.

Rules fixed in advance. Before the first confirmatory relation was evaluated, we wrote the following rules into the design record of the project, judged per ground truth over the 23 relations with a budget of 1,000 answers, an uncertified run counting as 1,000, with the two classes reported separately. (V1) Paired by relation, the median ratio of the answers of Cert3NF to those of the two phases with the LLM is at most 0.9, and Cert3NF needs fewer answers in total. (V2) Every certificate of Cert3NF is correct: an independent checker accepts it, every premise is right for the ground truth, and its design is good, except on the relations listed before the run where the data violate a documented FD. (V3) Paired by relation, the median ratio of the answers of Cert3NF to those of the EC^2 order, with and without the LLM and each the median of three seeds, is at most 1. (V4) The ratio of answers to the price of Thm. 4.4 has a median of at most 1.6 and an upper quartile of at most 2, and no certificate has fewer no answers than m_r. (V5) Cert3NF certifies at least as many relations as without an advisor, needs at most b answers more than it, plus one per maximal set answered yes and per wrong first guess, on at least 90% of the relations, and has at least as many good designs as the posterior certificate at the end of a budget of 100 answers, counted per relation by the median of its three seeds. The rules were judged once. V1, V2, V3 and V5 hold on all relations, on class A and on class B under both truths. V4 holds on all relations and on both classes under both truths, except that the upper quartile on class B under the full truth is 2.03, by the $(n + 1)p$ rule for quantiles of the evaluation script.

Price and baselines. We find a smallest F exactly, by integer programming over the left-reduced FDs of Σ. Where the data violate a documented FD, F is a smallest cover of the FDs of Σ^+ that hold on the data, and the held sets are the maximal sets of Σ that contain the left-hand side of a false candidate. These are also the held sets of the FDs of Σ^+ that hold on the data, so the price is that of these FDs. The automatic methods, the normalizer, both forms of Normalize, the second LLM, attribute exploration and EC^2 ran after these verdicts or apart from them, and no rule depends on them. The posterior certificate and EC^2 draw 80 samples from the posterior, and the posterior certificate accepts a design at $\delta = 0.1$. EC^2 weighs the samples of the posterior equally, assigns each sample to the first design that covers it in a greedy cover of the samples by good designs, and asks per round about the five candidates of largest expected weight of cut edges, with ties broken by the largest marginal. More samples or smaller rounds hardly help EC^2: with 320 samples it certifies 4 (full) and 7 (sure) of the 23 relations within 100 answers, and with one question per round 5 and 8, against 5 and 7 above, with at most one more good design and no wrong certificate. Without the answers of the LLM, EC^2 and the posterior certificate certify at most 4 (full) and 5 (sure) of the 23 relations within 100 answers, and obtain 15 and 21 good designs, against 18 and 23 of Cert3NF.

Sums of answers. Sums of answers over the 23 confirmatory relations count an uncertified run as 1,000 answers, which favours the EC^2 order on hgnc. Without a budget, Cert3NF spends 897 (full) and 222 (sure) answers above the price, its variant without an advisor 2,264 and 923, the EC^2 order 1,800 and 682, and the EC^2 order with the LLM 2,865 and 1,488. Finding the FD behind a yes takes 862 and 117 of these answers of Cert3NF, and 1,295 and 533 of those of the EC^2 order. Paired by relation, Cert3NF needs fewer answers than the EC^2 order and the two phases by the Wilcoxon signed-rank test ($p \leq 0.003$ under either truth), with 95% bootstrap intervals of the median ratio of $[0.62, 0.91]$ (full) and $[0.56, 0.74]$ (sure) against the EC^2 order and $[0.76, 1.00]$ and $[0.73, 0.95]$ against the two phases. The LLM changes the answers of Cert3NF by no significant amount ($p = 0.61$ and 0.25), and resampling the twelve publishers instead of the relations moves no end of these intervals by more than 0.11. The four relations without a true candidate cost exactly their price, and without them the median ratios of answers to the price are 1.44× and 1.28×, against 1.37× and 1.26× over all 23. On the development relations, Cert3NF certifies 31 (full) and 34 (sure) of 35 within the budget and all without one, at a median of 1.36× and 1.30× the price.

Normalize. We reimplemented Normalize as described [42], with the declared key as the primary key. Its score of an FD is the mean of four features that favour a short left-hand side with short values, adjacent attributes and many duplicate values, and Normalize order gives each candidate $X \to A$ the score of the FD from X to the attributes that the mined FDs derive from X, without X and the key. Everything else, including the recalibration by the designer's answers, is that of Cert3NF. Within 50 answers, Normalize order certifies 17 (full) and 20 (sure) of the 23 relations, against 16 and 19 of Cert3NF, and in total it needs 1,755 and 834 answers, against 1,774 and 851. Paired by relation, the median ratio of the answers of Cert3NF to those of Normalize order is 0.97 and 0.96, which the Wilcoxon signed-rank test does not distinguish from 1 ($p = 0.94$ and 0.19), and no certificate is wrong.

A second LLM. With gpt-oss-20b, which errs on 29% of the confirmatory candidates while Qwen errs on 41%, Cert3NF certifies as many relations within the budget as with Qwen. Indeed, where one LLM errs, the odds that the other errs are 5.4× higher, so the better of the two lowers the error rate but leaves the hard candidates to the designer.

Table 9: Answers until the Certificate on the Development Relations, as in Tab. 3; ✗ Marks a Wrong Certificate

		full truth						sure truth					
relation	cand.	true	price	Cert3NF	LLM	w/o adv.	post.	true	price	Cert3NF	LLM	w/o adv.	post.
food	78	31	33	64†	63†	(152†)	75†	7	11	33	33	49	55✗
hospital	145	54	36	67	87	(121)	96✗	22	34	70	66	99	80✗
pricepaid	34	13	15	22†	24†	55†	32†	4	7	12	19	26	33
ssb-customer	24	12	11	15	16	25	18	10	10	11	14	21	19
ssb-date	209	205	23	26	26	(154)	40	113	30	43	42	(118)	70
ssb-lineorder	3,814	13	44	51	53	86	–	5	33	37	37	48	85
ssb-part	86	3	11	12	14	20	35	3	11	14	14	20	40
ssb-supplier	21	11	10	13	16	22	15	9	9	10	13	20	15
tpcds-core	48	23	17	27	26	78	25✗	13	25	67	60	80	23✗
tpch-lineitem	36,883	10	49	54	61	96	59✗	10	49	55	68	96	–
tpch-orders	484	44	32	58	39	(151)	–	44	32	56	42	(151)	–
tpch-partsupp	2,405	25	36	48	44	(108)	–	25	36	52	44	(108)	–
ldbc-comment	1,784	26	40	50	50	(208)	–	26	40	49	49	(208)	–
ldbc-study	1,249	39	46	67	65	(212)	–	39	46	71	65	(212)	–
gtfs	1,170	23	50	79	76	(167)	–	23	50	60	67	(167)	–
faa	695	14	38	55	53	98	80	14	38	55	60	98	95
abs	78	59	27	29	31	(156)	73	59	27	29	30	(156)	67
sec	428	16	35	45	46	(103)	92	15	37	55	55	(109)	100✗
scorecard	5,169	64	62	95	(103)	(161)	–	33	66	86	79	(152)	–
epa	1,895	25	124	(170)	(213)	(202)	–	13	125	(188)	(206)	(194)	–
pums	2,056	820	278	(317†)	(332†)	–	–	8	26	31	31	61	–
chicago	314	21	32	50	50	(115)	100	13	26	40	42	74	95
usda	161	69	41	51	56	(154)	97	25	31	40	40	(125)	70
auctionmark	4,243	36	33	41	41	(114)	–	15	47	54	52	(125)	98
seats	4,640	292	54	(101†)	(102†)	(345†)	–	19	40	57	57	(122)	–
tatp	909	28	38	50	49	81	–	28	38	47	47	81	–
simplewiki	139	36	44	72†	71†	(121†)	–	30	40	48	54	80	100
chembl	385	204	54	100†	(103†)	(674†)	–	43	29	39	44	(156)	–
musicbrainz	85	71	35	47†	50†	(170†)	80†	61	34	46	46	(137)	73
pluto	1,572	202	108	(151†)	(211†)	(488†)	–	46	42	56	55	(111)	–
fhrs	1,488	25	30	34	36	70	–	25	30	34	34	70	–
geonames	2,900	12	43	77	96	(144)	–	12	43	73	75	(144)	–
eia-860	1,097	14	35	43	48	100	–	14	35	40	46	100	–
stats19	1,670	42	33	45	45	79	60	42	33	42	45	79	55
ghcn-daily	351	13	28	37	37	80	98	13	28	36	36	80	80
within 100				31/35	29/35	13/35	17/35			34/35	34/35	18/35	19/35

Nine designers. The experts of the study are database researchers and administrators from outside our group, and its practitioners are graduate students and developers who have taken a database course and designed schemata. The experts studied the documentation before each session, and half of the practitioners read it for ten minutes. Every person worked on fec-cand, syn-claims and tpcc-customer in three sessions on one day or two consecutive days, at least 15 minutes apart, in the orders of a Latin square, so that in every condition each relation comes first, second and third once. A session starts with a fixed block of six questions about candidates, six about maximal sets and six about other subsets, half of them true. It then runs Cert3NF in the careful mode with $\theta = 0$ until the certificate, the end of the method or 100 answers, and the designer may take breaks, which do not count as answering time. Since maximal sets are usually large, a question about a maximal set names the attributes outside it: must any two records that agree on all attributes except B_1, B_2 and A also agree on A? Such a question is therefore short, and in the runs of (Q2) it names a median of four (full) and three (sure) attributes, as many as the median question of Cert3NF. The registered claim that maximal sets are as answerable as candidates, with at most five points more errors at the upper end of the 95% interval, is consistent with all three conditions, which erred on 0%, 2% and 2% of the maximal sets against 0%, 2% and 2% of the candidates, although 54 answers per form leave intervals of more than five points. The registered claim that errors are mostly independent, with the upper end of the 95% interval of $\hat{\phi}$ below 0.25 for the experts and for the practitioners with the documentation, is consistent with the latter, whose $\hat{\phi}$, estimated from the first two answers to each repeated premise, is 0.12, but three people per condition do not establish it, and the experts gave no wrong answer in the sessions from which to estimate it. An answer of

don't know counts towards the budget, is not an error, and leaves its candidate or maximal set undecided.

Attribute exploration. We run exploration by NextClosure [15] on the confirmatory relations, with the attributes in their order in the relation, the agree sets of the data as the known objects, Σ_0 as background knowledge, and the simulated designer as the expert. The classical expert refutes an implication by the smallest counterexample, the closure of its premise under Σ, and we also let it return the largest, a maximal set of Σ that contains the premise. To compare with Cert3NF, an implication becomes one yes or no answer per attribute of its conclusion that the confirmed implications do not yet imply, and the largest counterexample one answer per attribute by which it is grown, unless the data or the confirmed implications decide it. With the smallest counterexamples, exploration poses from 9 to almost three million implications, and eight relations under the full truth and nine under the sure truth do not finish within two hours. With the largest, it poses a median of 22 implications, fewer than the price, since one yes confirms a whole conclusion and one counterexample is a maximal set, and Sec. 6.3 reports the answers they amount to.

Synthetic relations. To see whether the answers follow the price beyond the 25 columns of the real relations, we generated denormalized joins of snowflake schemata whose true FDs hold by construction. The fact key (order, line) is declared, the entities form a tree under the fact, each with a key, one to four descriptive attributes and, with probability 0.3, a name that is a second key, and the remaining attributes are measures that only the fact key determines. Cardinalities and domains are drawn log-uniformly, values uniformly, and the attributes are renamed in a random order. One sweep varies the number of attributes at a share of 0.6 of entity attributes among those outside the fact key, and the other varies the share at 30 attributes, each with three seeds of 20,000 rows. Before generating any data, we fixed that Cert3NF and its variant without an advisor certify every design within 1,000 answers (S1), that the median ratio of answers to the price is at most 2 in every configuration (S2), and that Spearman's rank correlation of the answers is larger with the price than with the number of candidates (S3). All three hold (Tab. 5 in Sec. 6.3).